

RAIN PREDICTION SYSTEM

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INTRODUCTION

RAIN PREDICTION CHALLENGE



THE MISSION — BUILDING A MACHINE LEARNING PIPELINE TO PREDICT WHETHER IT WILL RAIN THE NEXT DAY IN AUSTRALIA, BASED ON HISTORICAL WEATHER MEASUREMENTS.

PROJECT OBJECTIVE

- Train, compare and evaluate multiple predictive models on real weather data, then interpret the results to understand what primary factors drive rain predictions.

KEY CHALLENGES

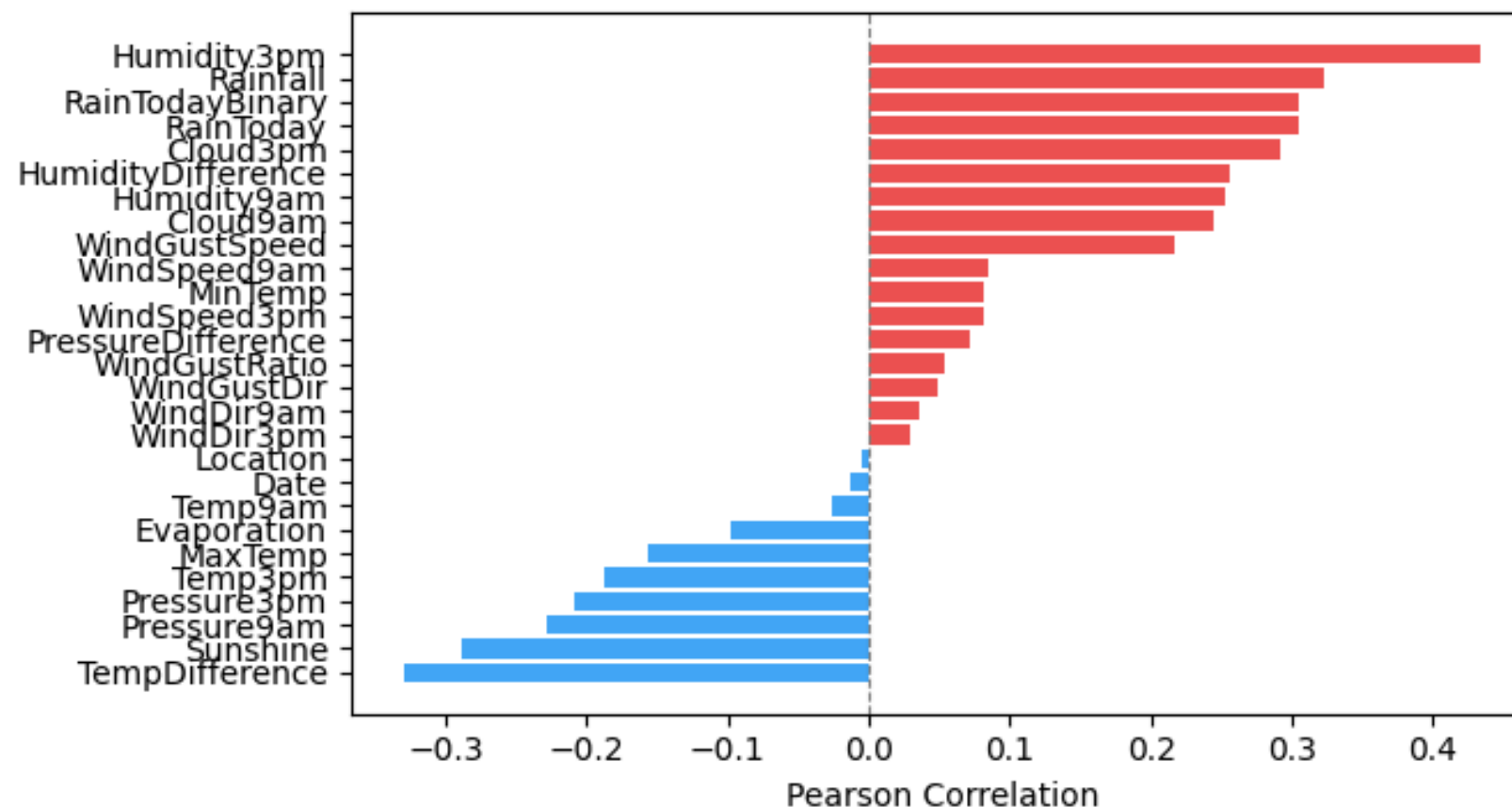
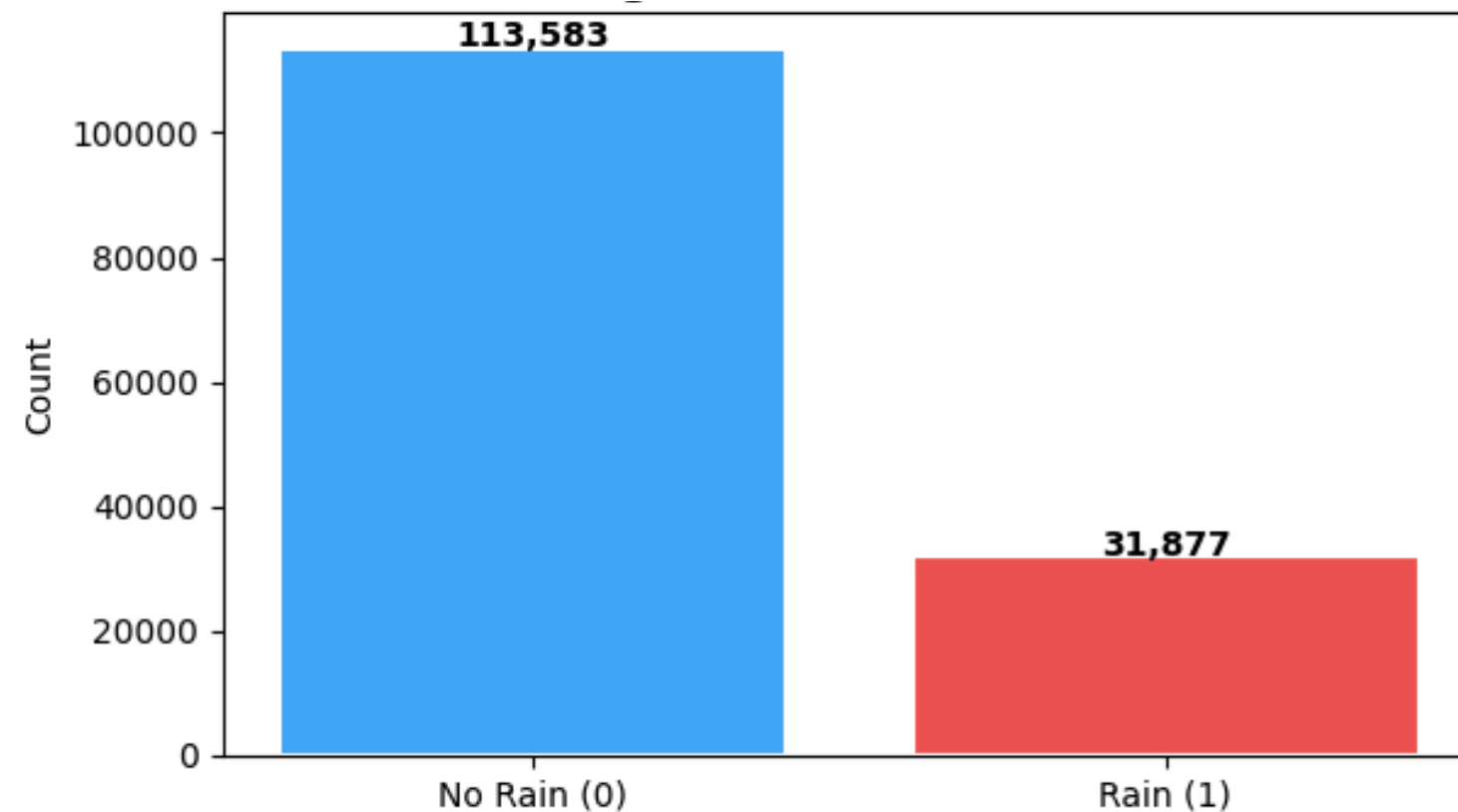
- Extreme Imbalance: 78% No Rain vs 22% Rain, so the model can't just "always say No Rain"
- Weather complexity: Many correlated features, missing values, and location-based patterns
- We need to balance recall (catching real rain days) vs precision (avoiding false alarms)

THE GOAL

- Predict Rain Tomorrow (yes/no) with the highest possible ROC-AUC
- Understand which weather variables matter most
- Construct a clean classification pipeline easily deployed for general public safety forecasts

DATASET

FEATURES & STRUCTURE



DATASET OVERVIEW

- Type: Tabular, binary classification
- Size: 145,460 daily observations (Kaggle)
- Target: RainTomorrow (0 = No Rain, 1 = Rain)

FEATURES

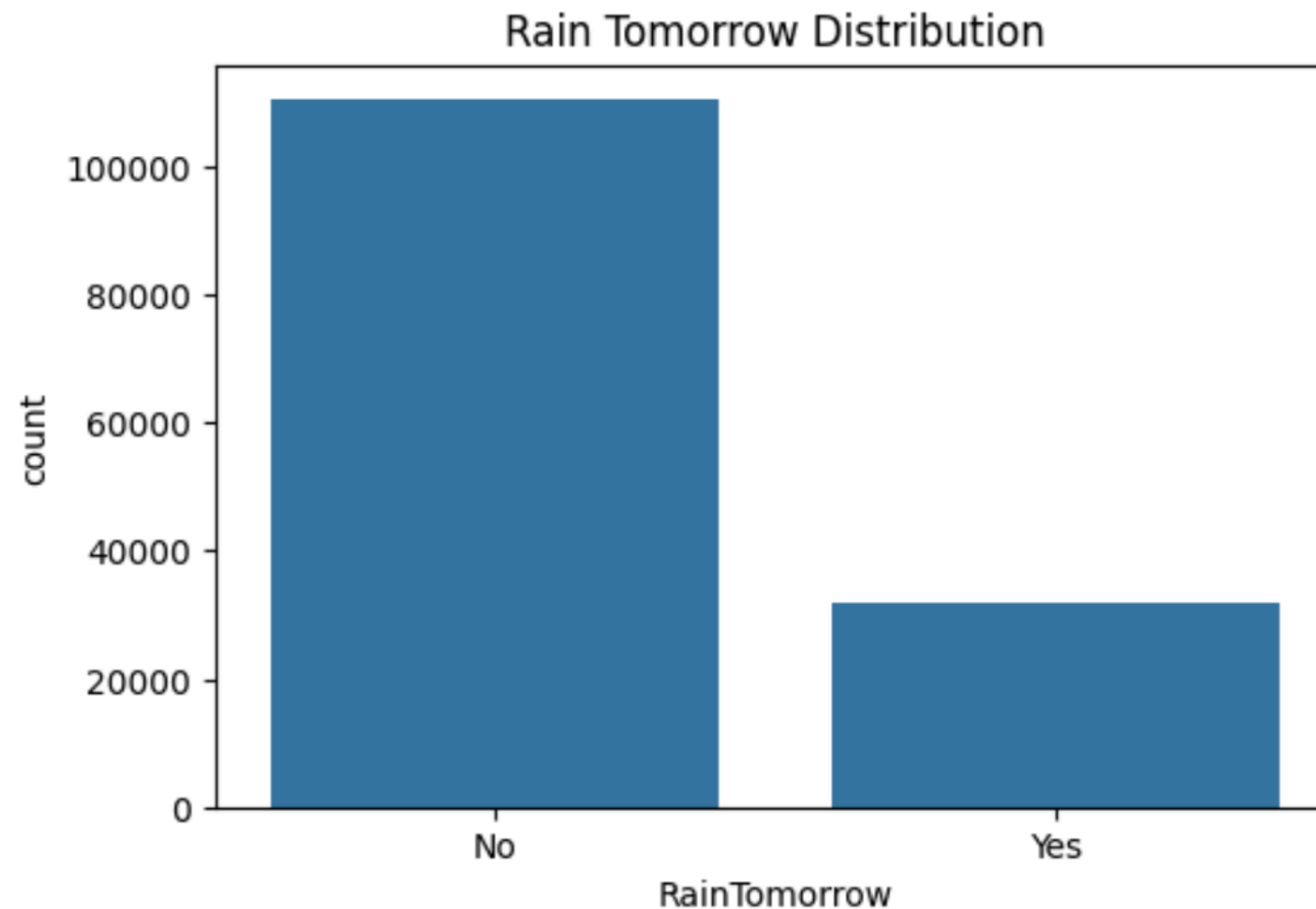
- Total: 25 features (20 Original + 5 Engineered)
- Engineered: TempDifference, HumidityDifference, PressureDifference, WindGustRatio, RainTodayBinary

TRAIN / TEST SPLIT

- Split: 80 / 20 Stratified (116,368 train / 29,092 test)
- Purpose: Stratification preserves the 3.56:1 class balance (78% No Rain vs. 22% Rain)

DATA EXPLORATION

PERFORM EXPLORATORY DATA ANALYSIS



INITIAL DATASET OVERVIEW

Exploratory analysis of over 140,000 daily weather records in Australia.

TARGET DISTRIBUTION

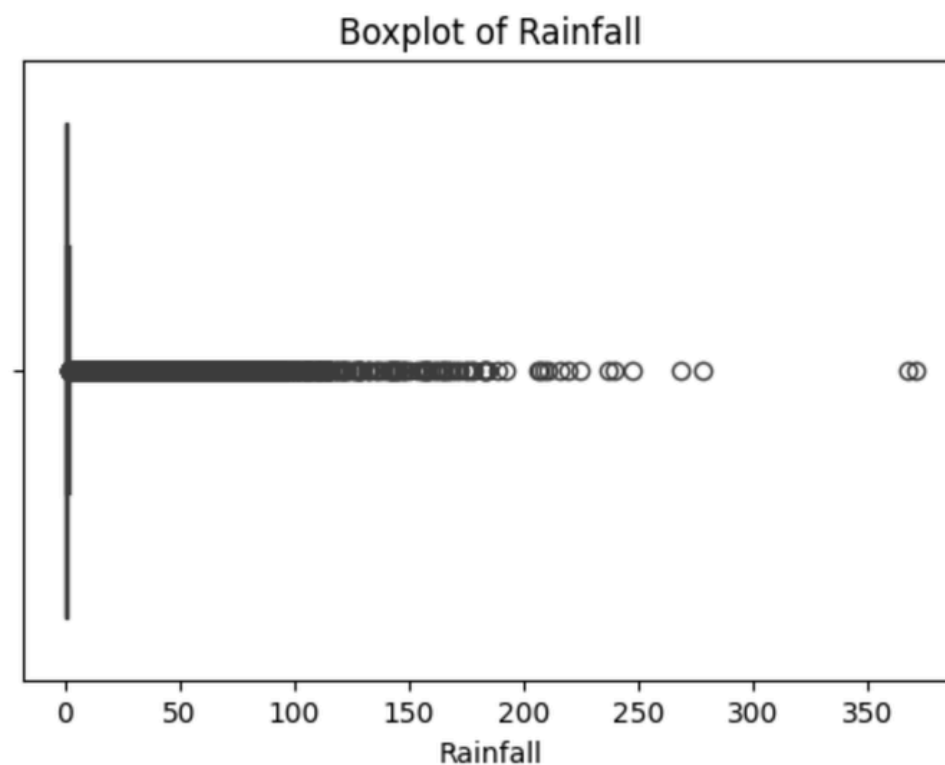
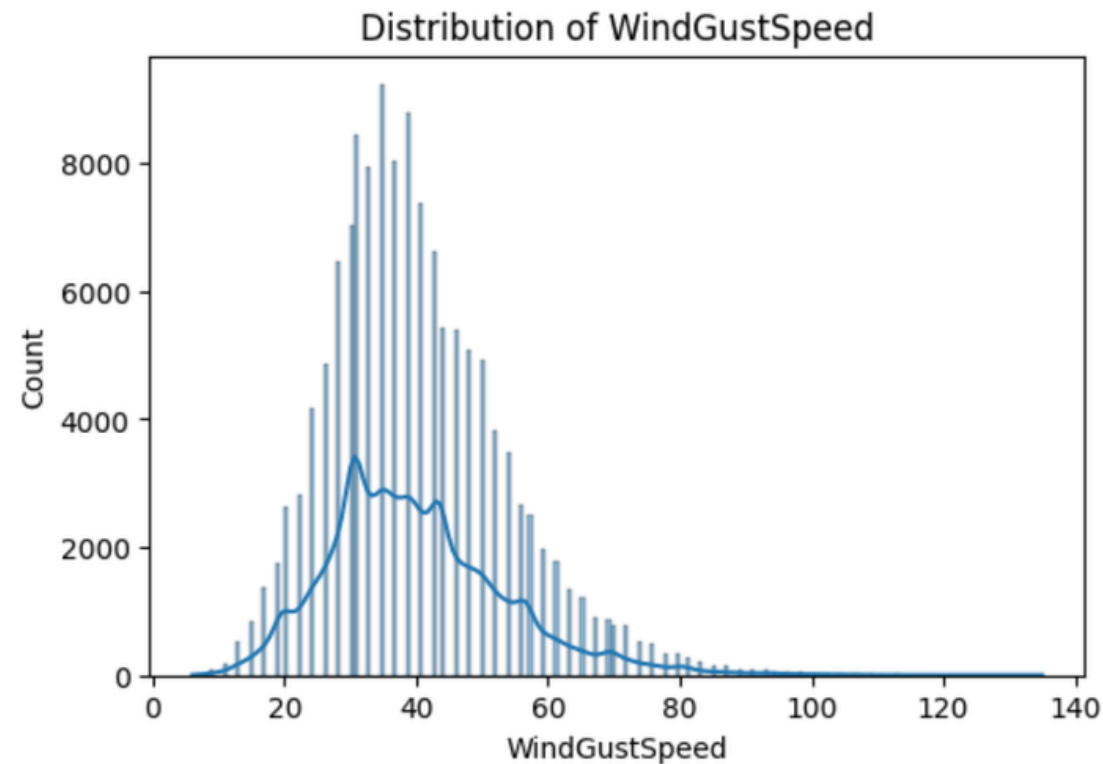
Severe Class Imbalance

~78% "No Rain"

~22% "Rain"

DATA EXPLORATION

VISUALIZE VARIABLE DISTRIBUTIONS; IDENTIFY OUTLIERS OR INCONSISTENT VALUES



VARIABLE DISTRIBUTIONS

Many meteorological features, such as wind gusts, exhibit a strong positive skew (long right tails).

OUTLIER DETECTION

Boxplots reveal a massive presence of extreme values (e.g., severe storms or sensor errors).

HANDLING INCONSISTENT VALUES

No dropping rows:

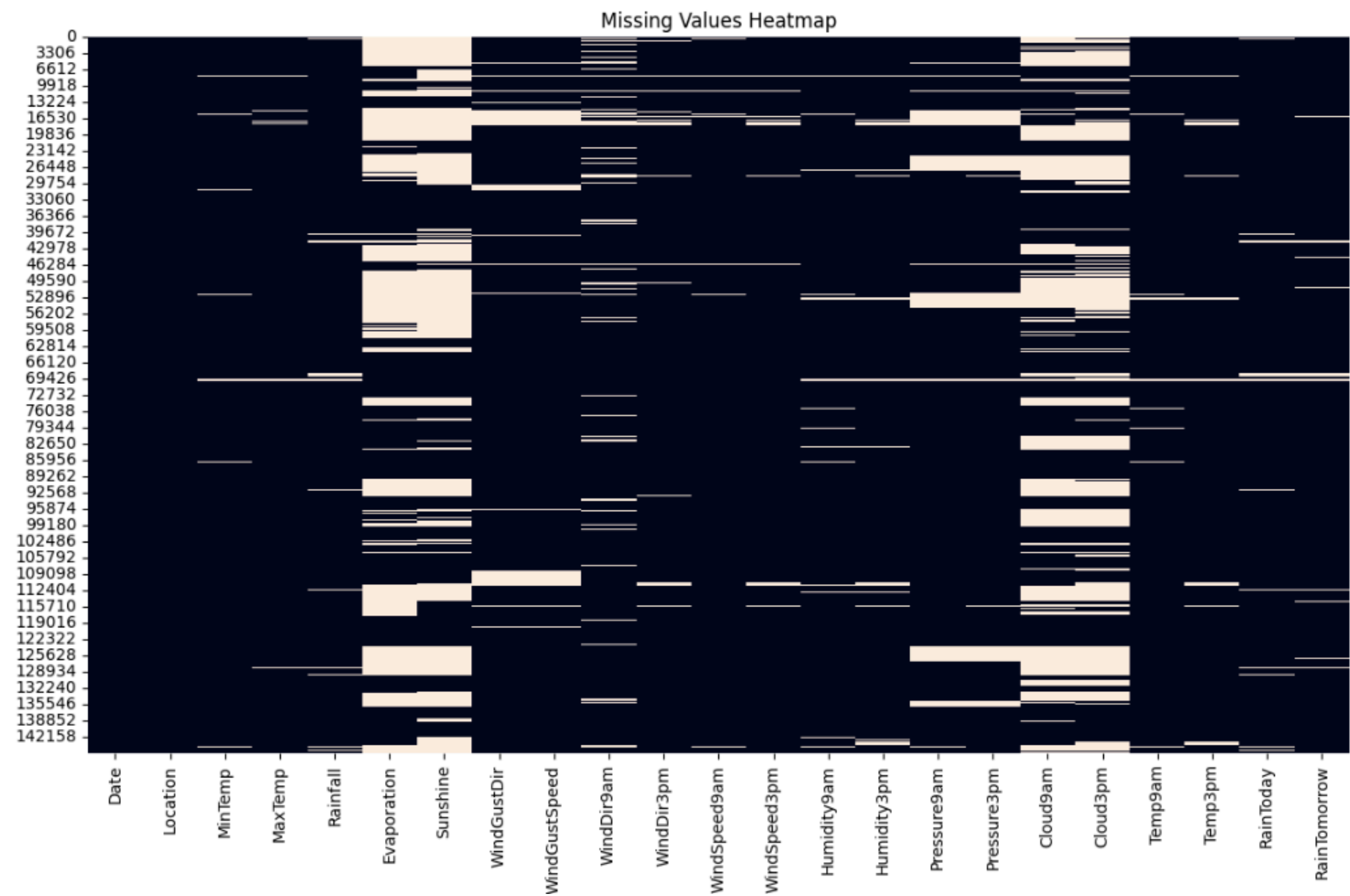
To prevent losing valuable historical data in an already imbalanced dataset.

IQR Capping

Applied IQR (Interquartile Range) Capping to mathematically "clip" and confine extreme spikes, stabilizing the data for predictive models.

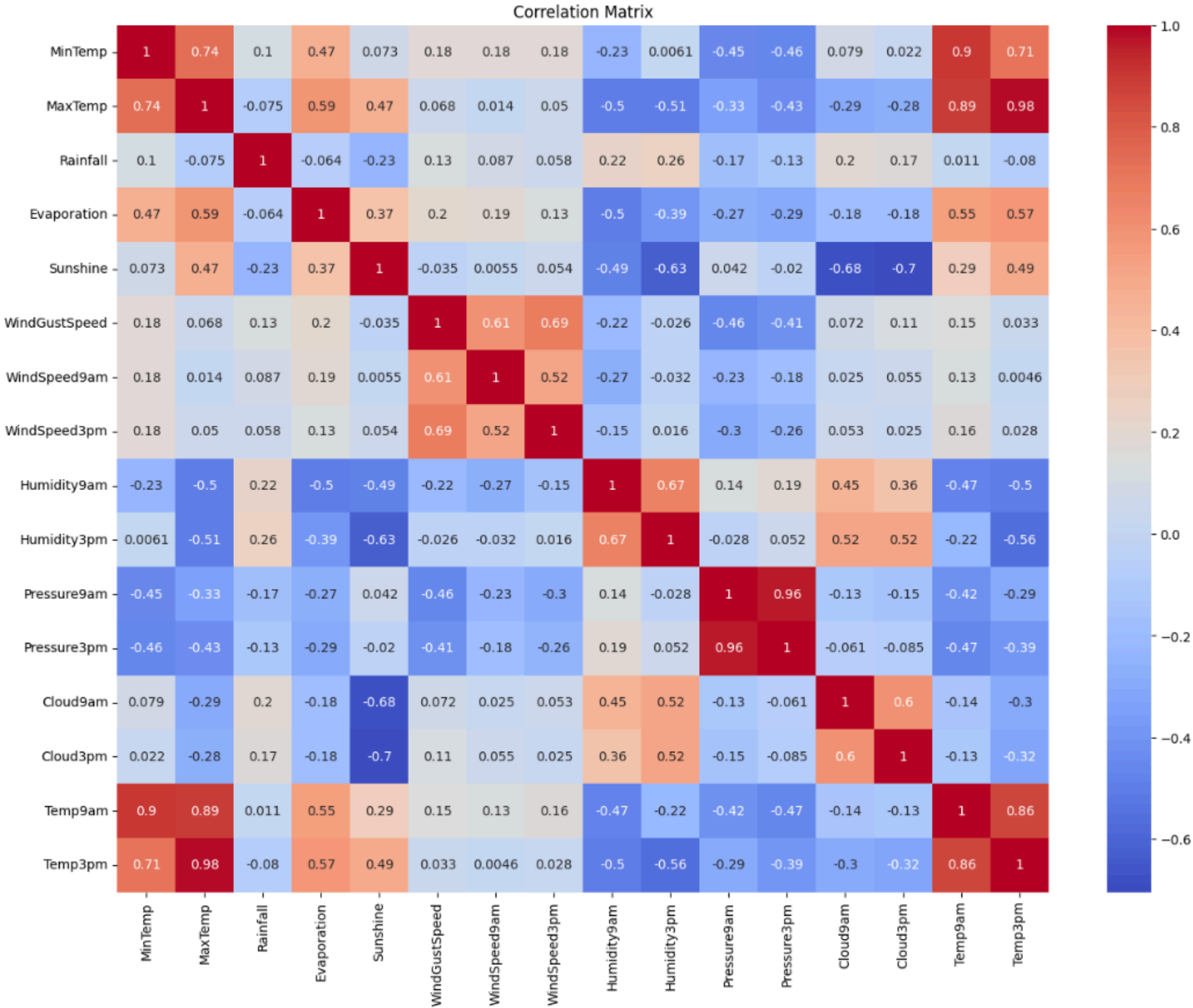
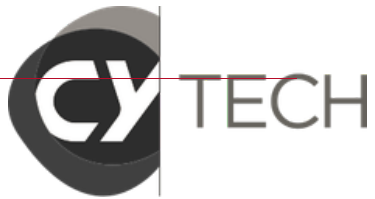
DATA EXPLORATION

DETECT MISSING VALUES; IMPUTE OR REMOVE MISSING DATA.



DATA EXPLORATION

ANALYZE CORRELATIONS BETWEEN VARIABLES



CORRELATION ANALYSIS

Positive Correlations:

Strong time-based links (e.g., Temp at 9am and Max Temp).

Negative Correlations:

Clear inverse relationship between Atmospheric Pressure and Humidity.

THE PREDICTIVE CHALLENGE

No single variable linearly explains the occurrence of rain.



This non-linear reality drove the creation of newly engineered features and the selection of predictive models.

FEATURE ENGINEERING

TRANSFORM NUMERICAL VARIABLES; ENCODE CATEGORICAL VARIABLES



CATEGORICAL ENCODING

BINARY MAPPING

Converted RainToday (Yes/No) into machine-readable binary formats (1/0).

LABEL ENCODING

Applied to complex categorical features (e.g., Wind Directions, Locations) to translate text into numerical vectors.

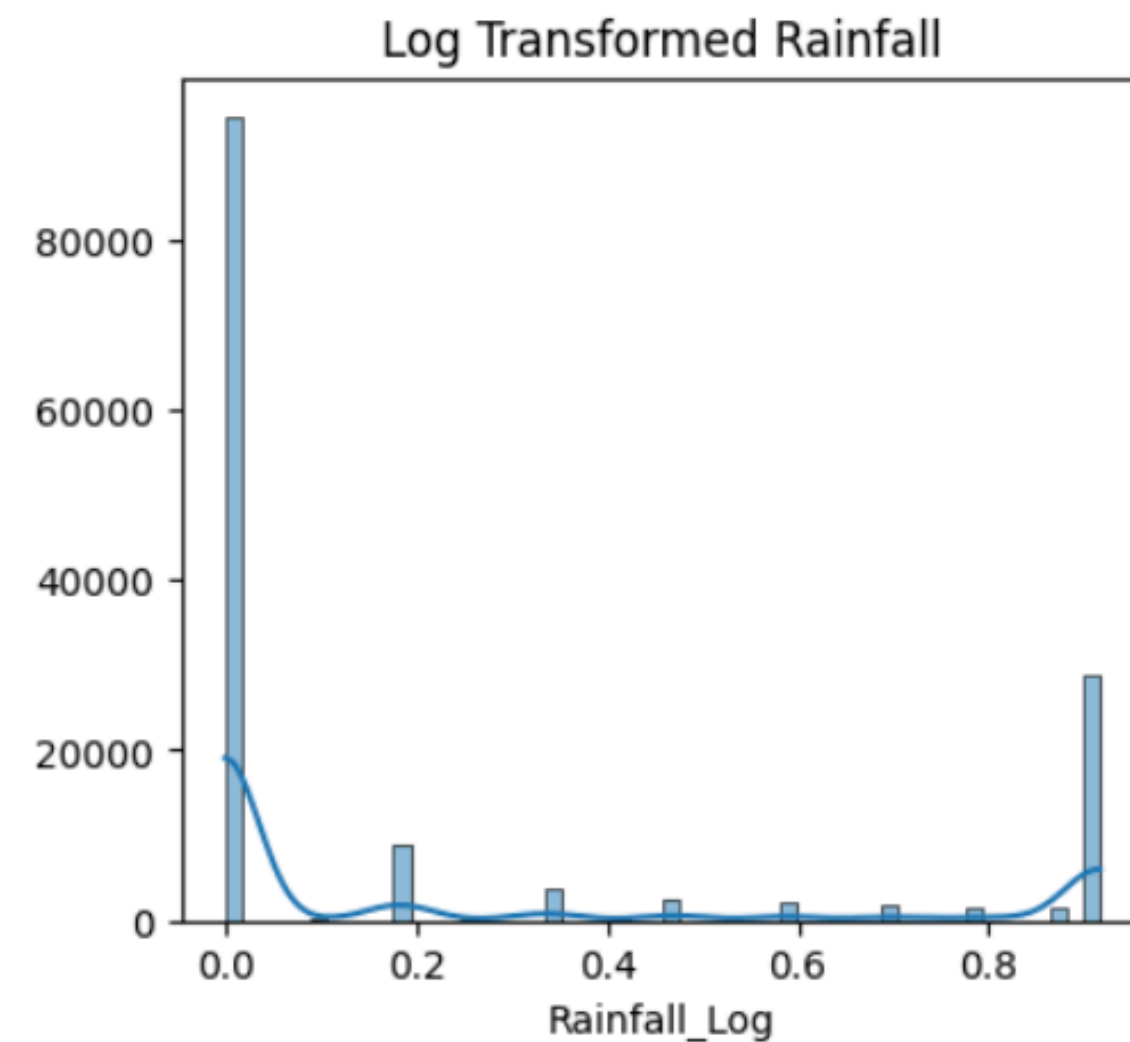
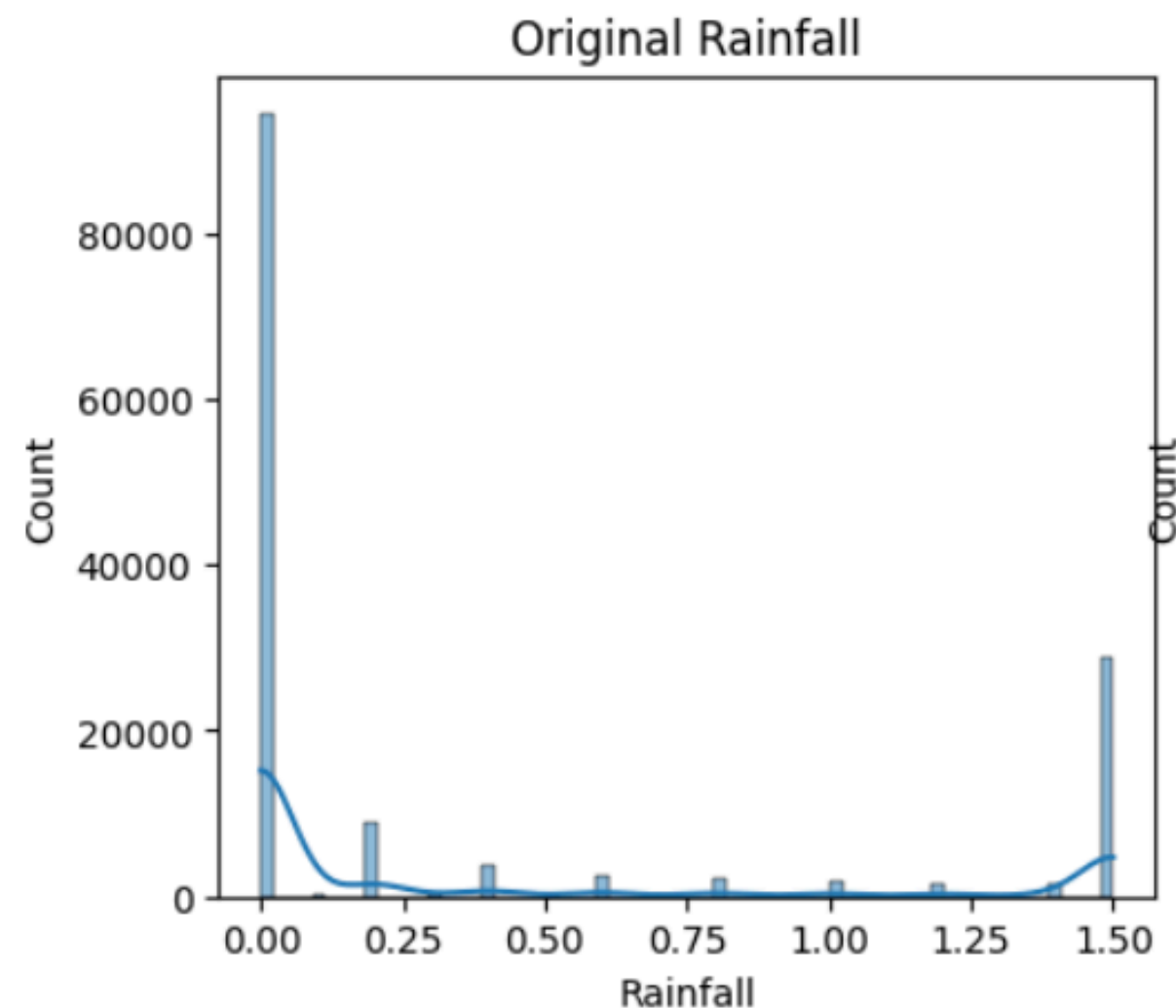
NUMERICAL TRANSFORM.

LOGARITHMIC TRANSFORM

Applied $np.log1p$ to the heavily right-skewed Rainfall variable.

JUSTIFICATION

Compresses extreme values while preserving the data distribution, stabilizing distance-based algorithms (like SVM) and neural networks (GRU)



FEATURE ENGINEERING

CREATE INTERACTION FEATURES; DERIVE NEW METEOROLOGICAL INDICATORS.



CAPTURING DAILY DYNAMICS

Created **TempDifference**, **HumidityDifference**, and **PressureDifference** (3pm - 9am).

JUSTIFICATION

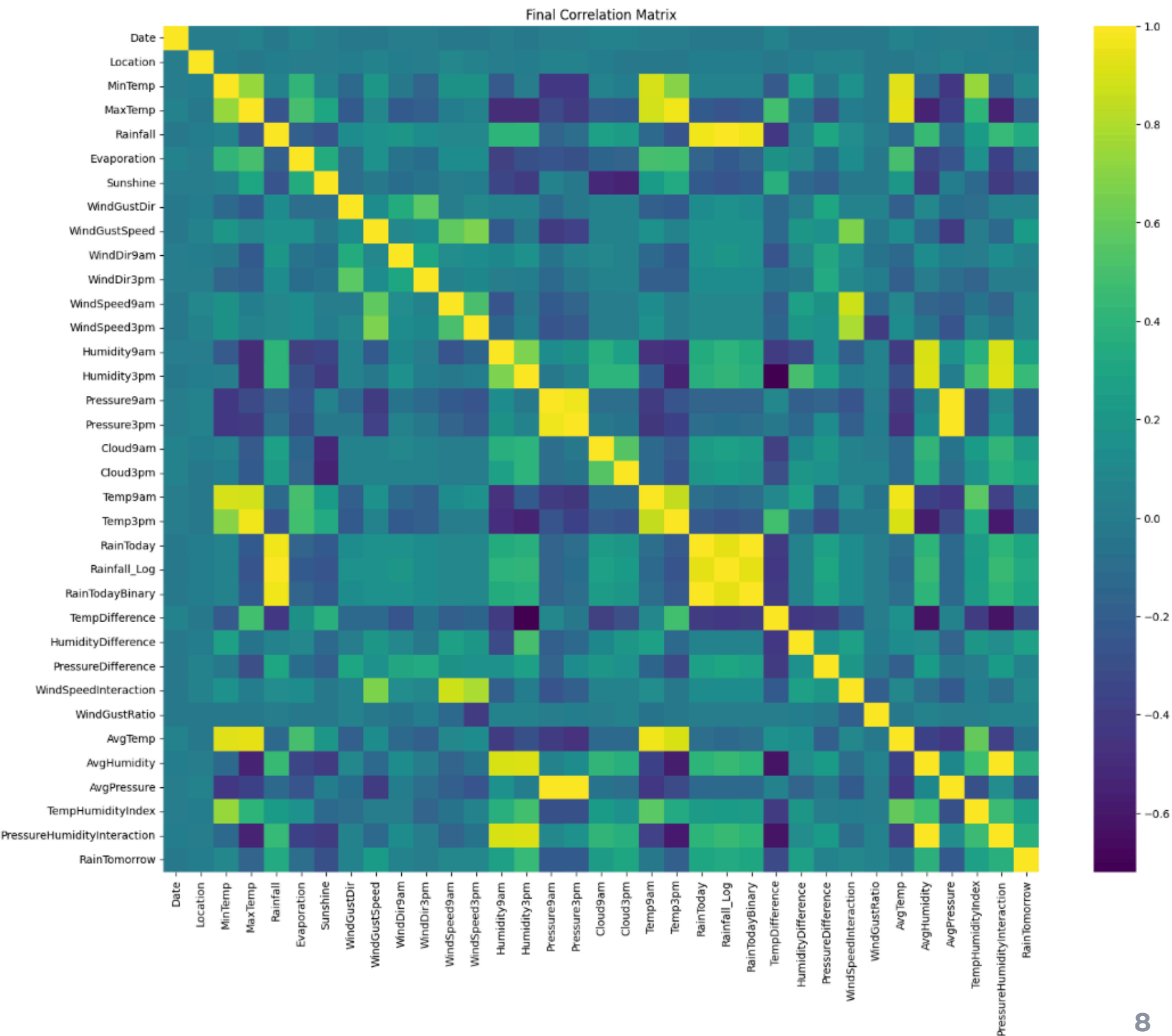
Weather is dynamic; a sharp drop in pressure is a stronger predictor of rain than the absolute pressure value.

COMPLEX METEOROLOGICAL INDICATORS

- WindGustRatio:** Highlights the intensity of gusts relative to baseline wind speed.
- PressureHumidityInteraction:** Combining average pressure and humidity into a single physical index.

JUSTIFICATION

Directly feeds the models with explicit physical relationships, significantly boosting predictive performance.



FEATURE ENGINEERING

NORMALIZE OR STANDARDIZE VARIABLES



THE SCALE PROBLEM

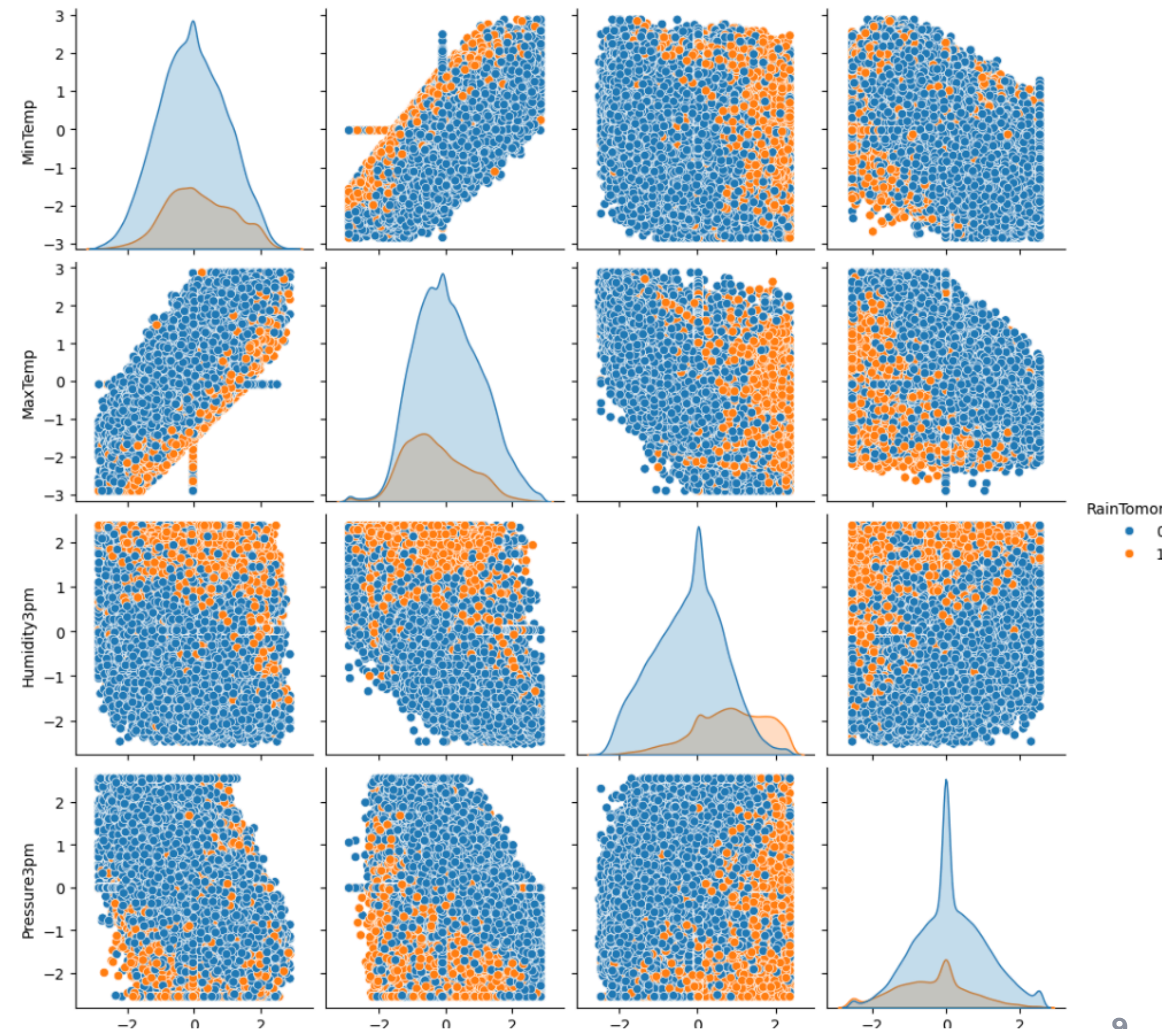
Features existed on completely different scales (e.g., Pressure > 1000 hPa vs. Temp around 20°C).

OUR SOLUTION (STANDARDIZATION)

Applied **StandardScaler** to all predictor variables.
Transformed data to have a mean of 0 and a variance of 1.

STRATEGIC JUSTIFICATION

Prevents variables with larger numerical ranges (like Pressure) from artificially dominating the loss function.
Absolutely strictly required for Gradient Boosting, SVM, and Deep Learning convergence.



EVALUATION METRICS

MEASURING REAL-WORLD IMPACT



Before looking at the results, how do we define success?

ACCURACY:

The overall percentage of correct weather forecasts.

WHAT IF IT'S LOW?

The model is generally **confused** and **randomly guessing** the weather.

PRECISION:

When our model predicts "Rain," how often does it actually rain?

WHAT IF IT'S LOW?

False Positives. We predict a storm, people cancel their outdoor plans or flights, but the sun is shining.

RECALL:

Out of all the actual rainy days, how many did we successfully catch?

WHAT IF IT'S LOW?

False Negatives. We predict sunshine, people leave their umbrellas at home, and they get caught in a downpour.

F1-SCORE:

The harmonic balance between Precision and Recall.

WHAT IF IT'S LOW?

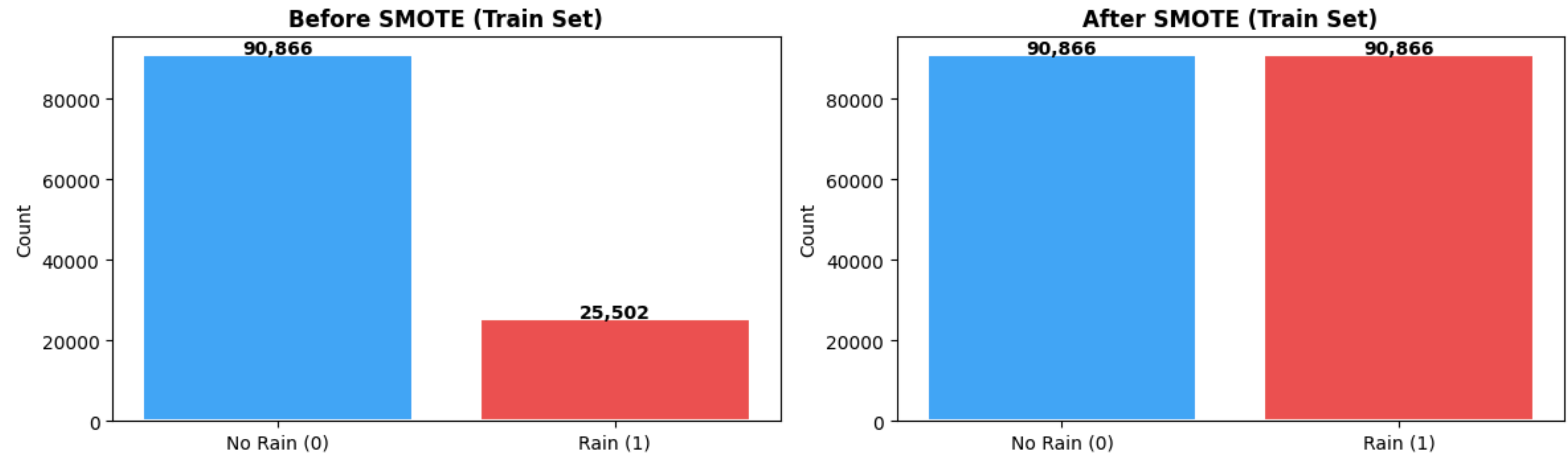
The model is heavily **biased**. For example, it might always predict "**No Rain**" just to artificially boost **accuracy**, ignoring the minority class.

SMOTE ANALYSIS

HANDLING CLASS IMBALANCE



Class Distribution — SMOTE Effect



RANDOM FOREST CLASSIFIER

THEORY, FUNCTION AND RELATION



WHAT IS A RANDOM FOREST?

ENSEMBLE LEARNING

Algorithm based on Decision trees that uses "majority voting" to aggregate predictions from hundreds of independent trees.

WHY WE CHOSE IT?

ROBUSTENESS

Its exceptional robustness against overfitting and weather outliers makes it the perfect standard baseline to evaluate and compare the performance of the other model.

HOW DOES IT WORK?

DECISION TREE CLASSIFIER

each decision tree is trained on a random subset of the data rather than the entire dataset. By combining the predictions of these diverse trees through a majority vote, the final model becomes highly accurate, stable, and resistant to overfitting.

CONNECTION TO OUR DATASET

CAPTURES NON-LINEARITY

this algorithm is ideal as it naturally identifies critical physical thresholds, such as specific humidity-pressure combinations that trigger rain.

HYPERPARAMETER TUNING

The Goal: Find the optimal model settings without guesswork.

The Method: The algorithm automatically tested random combinations of key parameters (number of trees, max depth, min number of split).

OVERFITTING ANALYSIS

The Results: Training Set (98.56%) vs. Testing Set (84.51%).

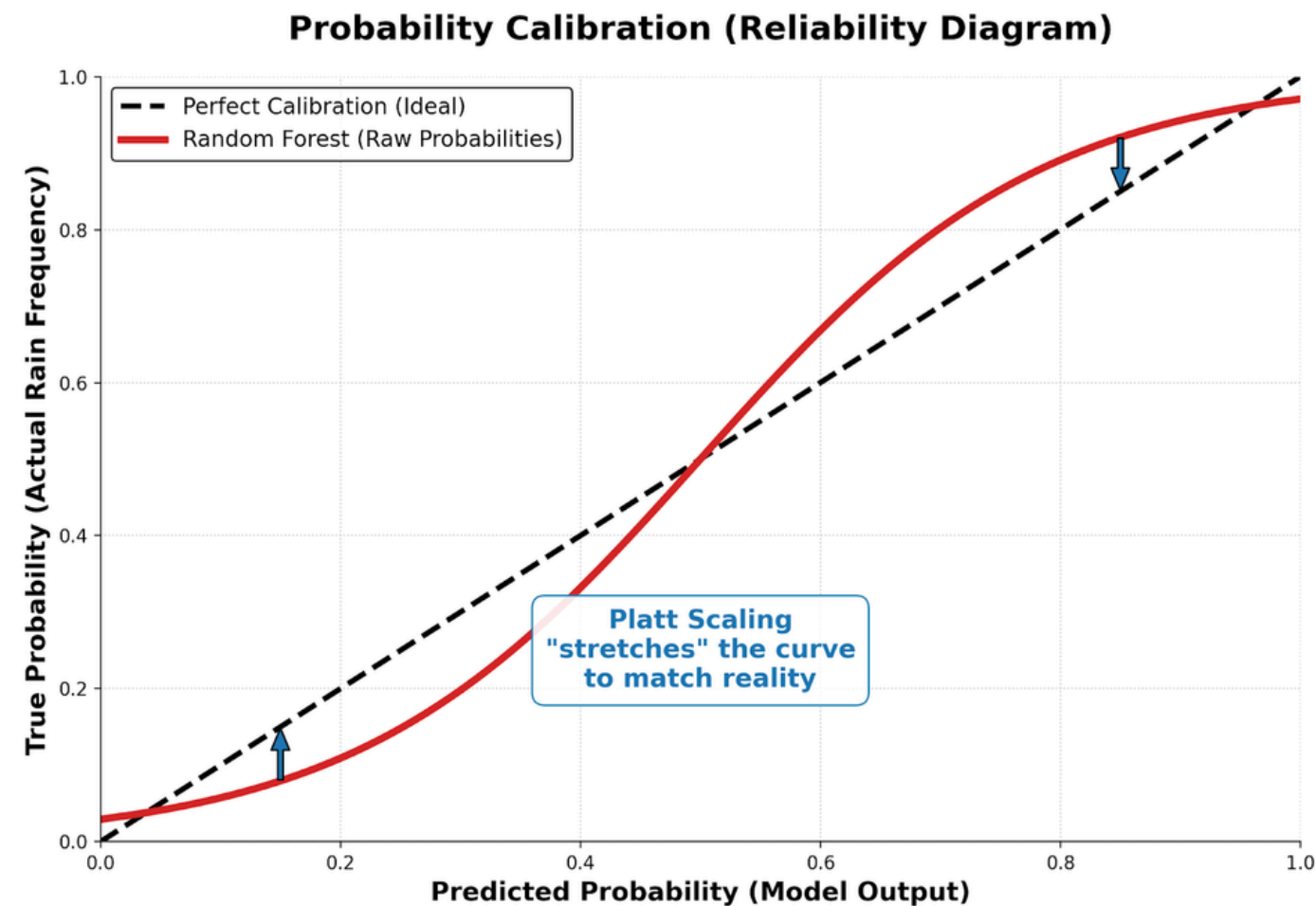
What it Means: The model partially "memorized" the historical data rather than perfectly generalizing the underlying weather patterns.

PROBABILITY CALIBRATION

The Problem: Because they average the votes of many trees, they struggle to predict extreme values (0 or 1), pushing probabilities toward the middle.

The Solution (Platt Scaling):

We applied `CalibratedClassifierCV` (sigmoid method) to mathematically correct this distortion.



Platt Scaling (Logistic / Sigmoid Function)

$$y = \frac{1}{1 + e^{-(A \cdot x + B)}}$$

A = Slope Parameter (Fixes probability compression)

B = Shift Parameter (Fixes class imbalance bias)

WHAT IS THE PROBLEM ?

Random Forests calculate probability by averaging the votes of 200 trees. Because every tree is different, they almost never agree 100%.

THE EFFECT

Probabilities get "squashed" toward the middle, forming an S-curve. The model struggles to reach absolute 0% or 100%.

SOLUTION: PLATT SCALING

We applied CalibratedClassifierCV (Sigmoid Method).

HOW IT WORKS

It compares the Forest's raw guesses with real weather data. It learns two mathematical parameters, A (slope) and B (shift), to correct the curve.

IMPACT

The Sigmoid formula recalibrates the distorted percentages (e.g., a raw 45% is mathematically fixed into a real 66%).

RESULTS

The model is now statistically reliable: an 80% output guarantees rain exactly 8 times out of 10.

RANDOM FOREST CLASSIFIER

RESULTS

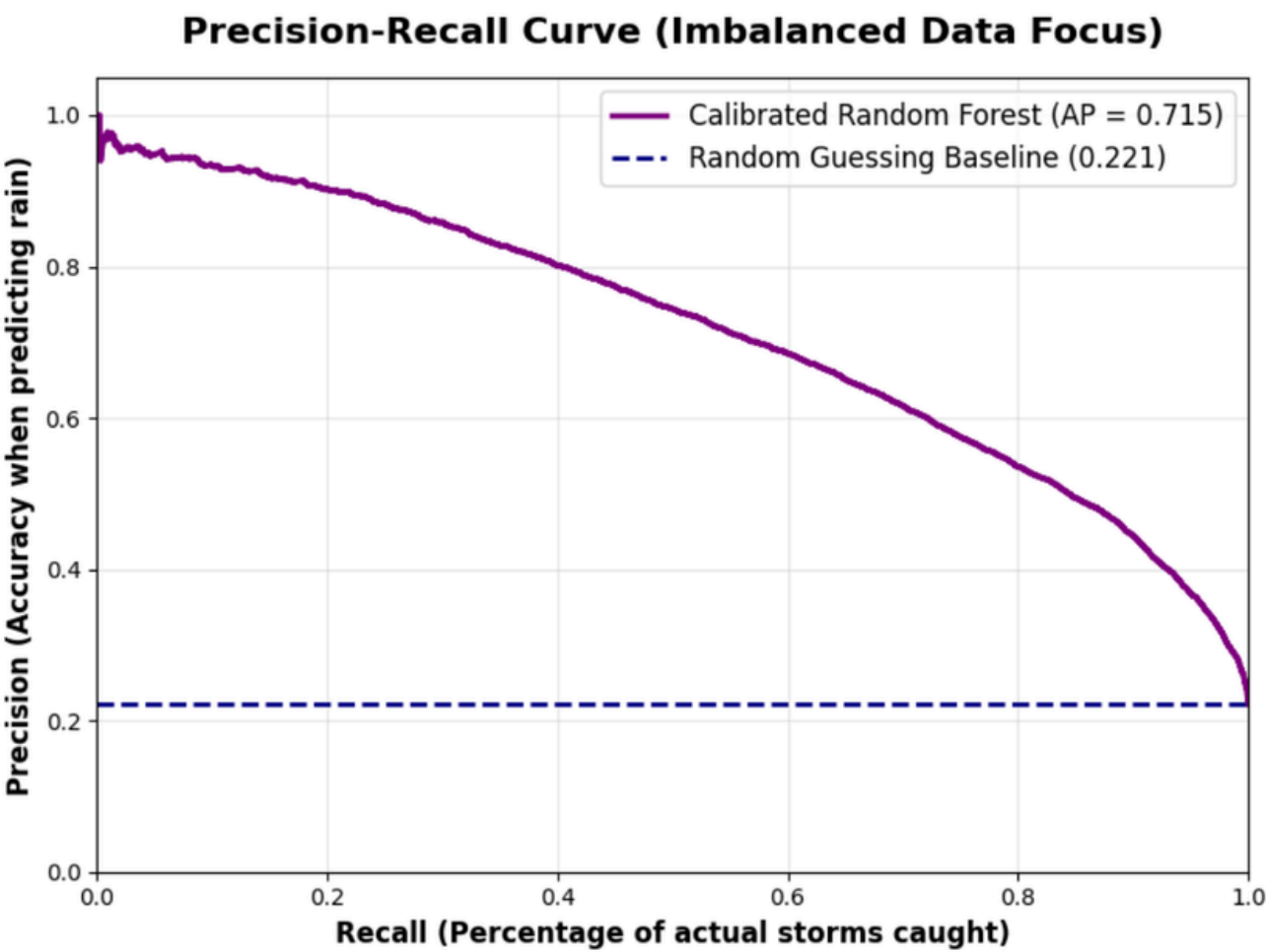
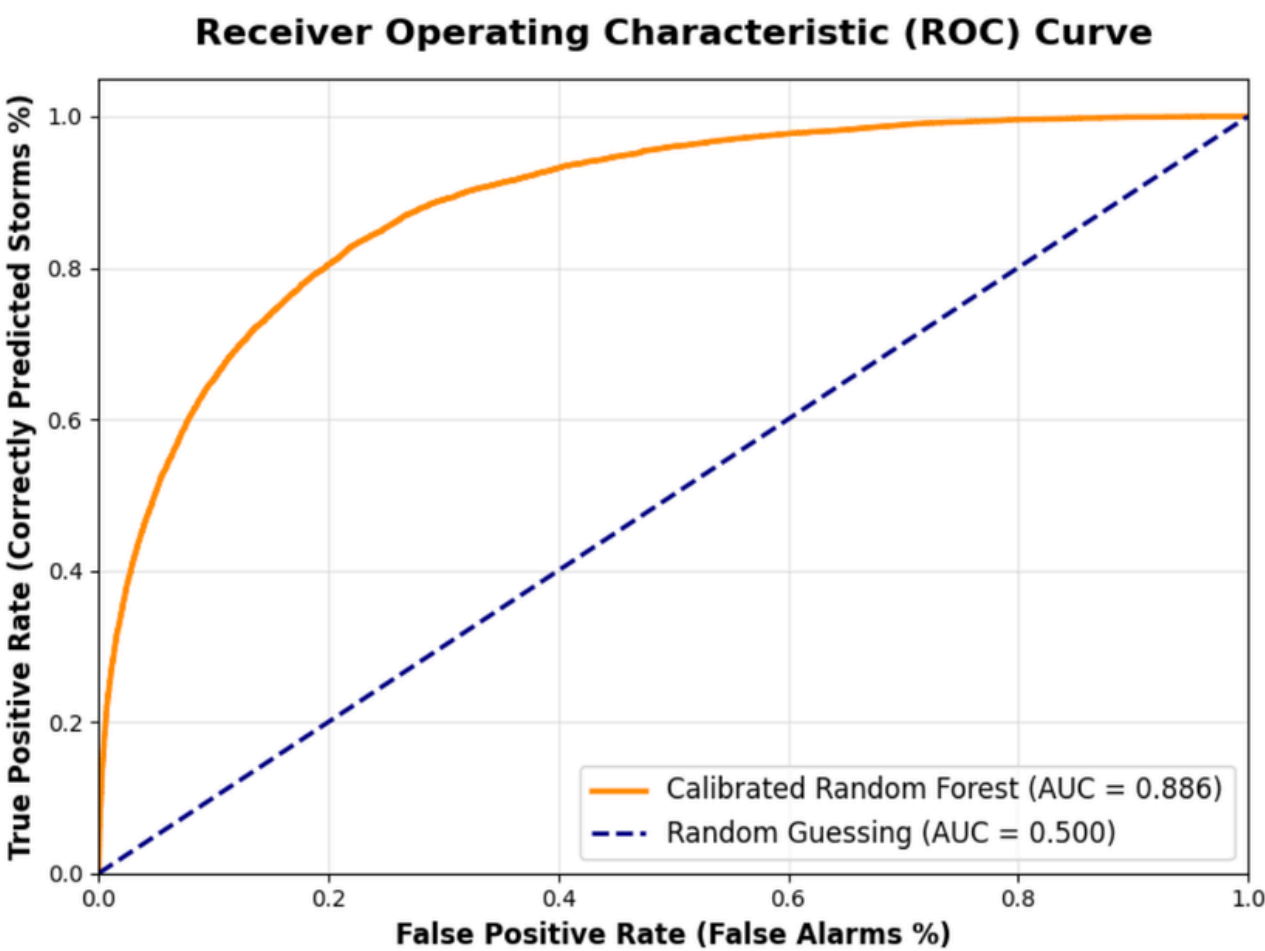


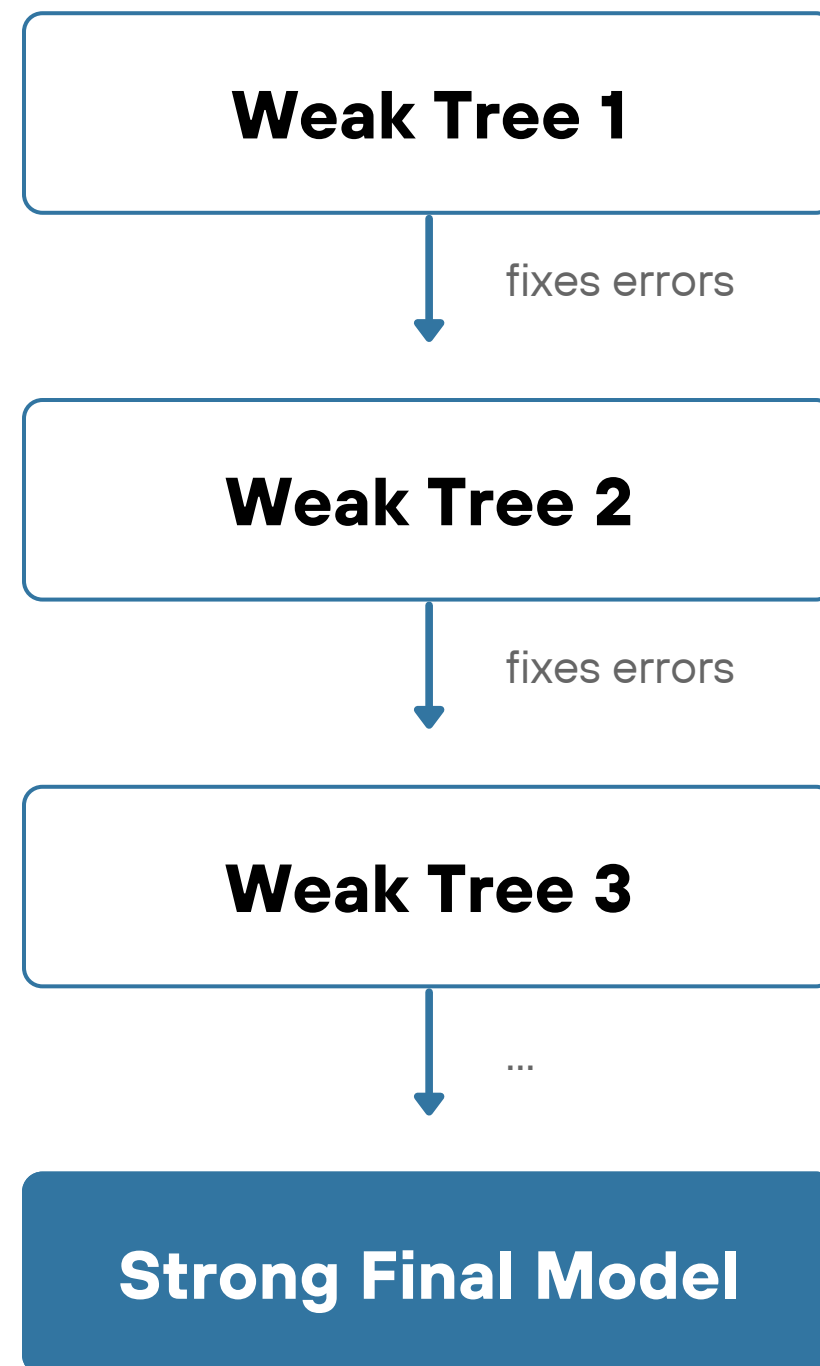
GLOBAL PERFORMANCE

	PRECISION	RECALL	F1-SCORE
0	0.88	0.94	0.91
1	0.72	0.53	0.61

SUPPORT
33987
9651

ACCURACY
85%





WHAT IS IT?

Gradient Boosting builds many small decision trees sequentially. Each new tree corrects the mistakes of the previous one. After 200 trees, they combine into one powerful predictor.

WHY WE CHOSE IT?

Weather prediction involves complex non-linear relationships between humidity, pressure, wind and temperature. Gradient boosting handles these naturally without assuming any linear relationship between features.

THE 3 MODELS USED

- Gradient Boosting (sklearn) — classic reference model
- XGBoost — faster, with regularization to prevent overfitting
- LightGBM — fastest, best for large datasets

KEY HYPERPARAMETERS

n_estimators=200

200 trees built sequentially

max_depth=5

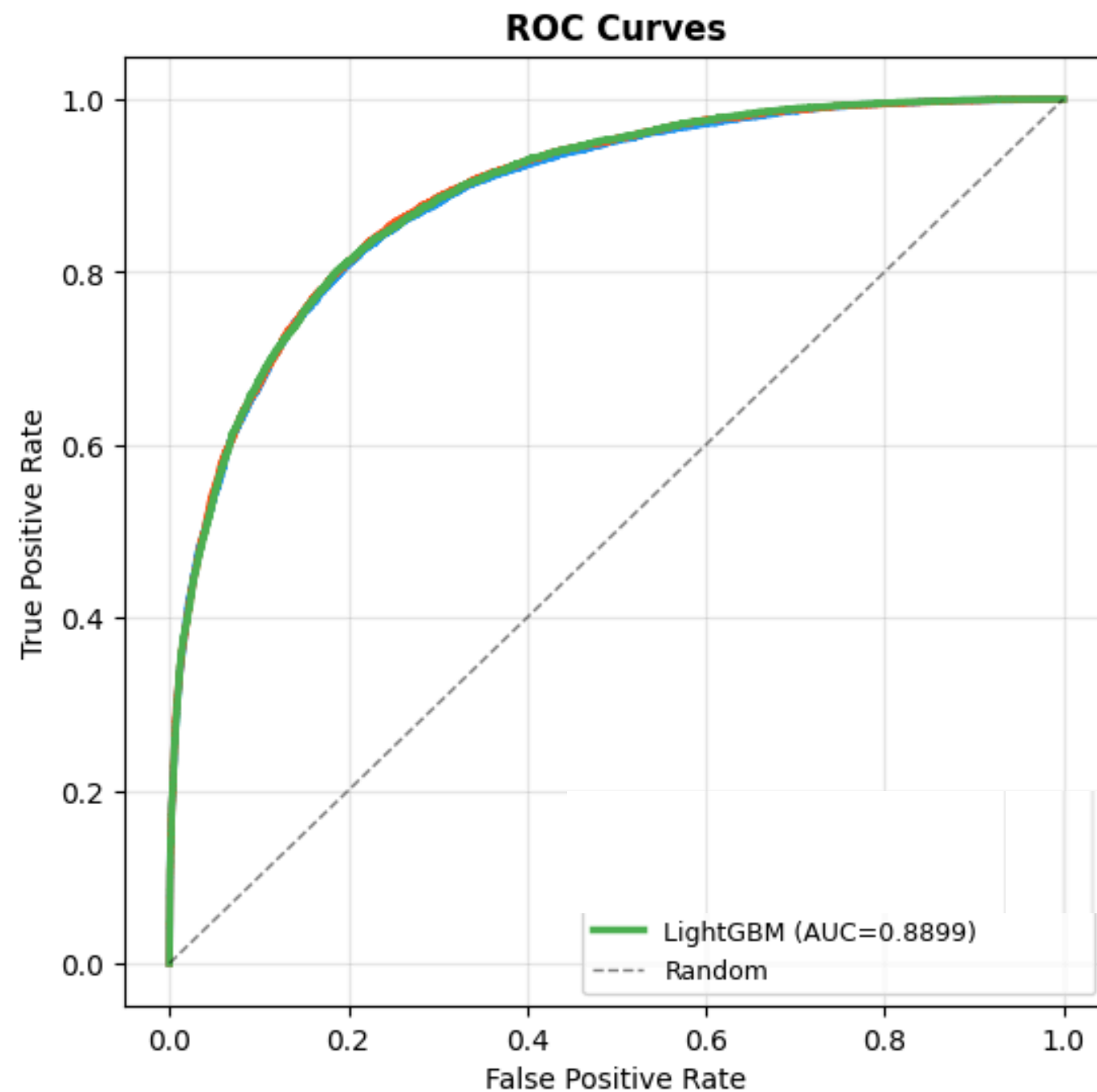
Each tree can have max 5 levels

learning_rate=0.1

Each tree contributes only 10%

subsample=0.8

Each tree sees 80% of data randomly



DATA PREPARATION

145,460 records cleaned, 25 features selected. 80/20 stratified train/test split → 116,368 train / 29,092 test samples.

KEY HYPERPARAMETERS

All 3 models trained with:

`n_estimators=200`, `max_depth=5`, `learning_rate=0.1`, `subsample=0.8`.
Regularization applied on XGBoost and LightGBM (`reg_alpha=0.1`, `reg_lambda=1.0`) to reduce overfitting.

HANDLING CLASS IMBALANCE

Ran two experiments, with and without **SMOTE**. SMOTE balanced the training set from 3.56:1 to 1:1 by generating synthetic rain-day samples. Test set kept untouched.

ENSEMBLE METHODS

Combined the 3 models using Soft Voting, Weighted Voting, and Stacking (meta-learner: Logistic Regression).

LIGHTGBM - GRADIENT BOOSTING RESULTS

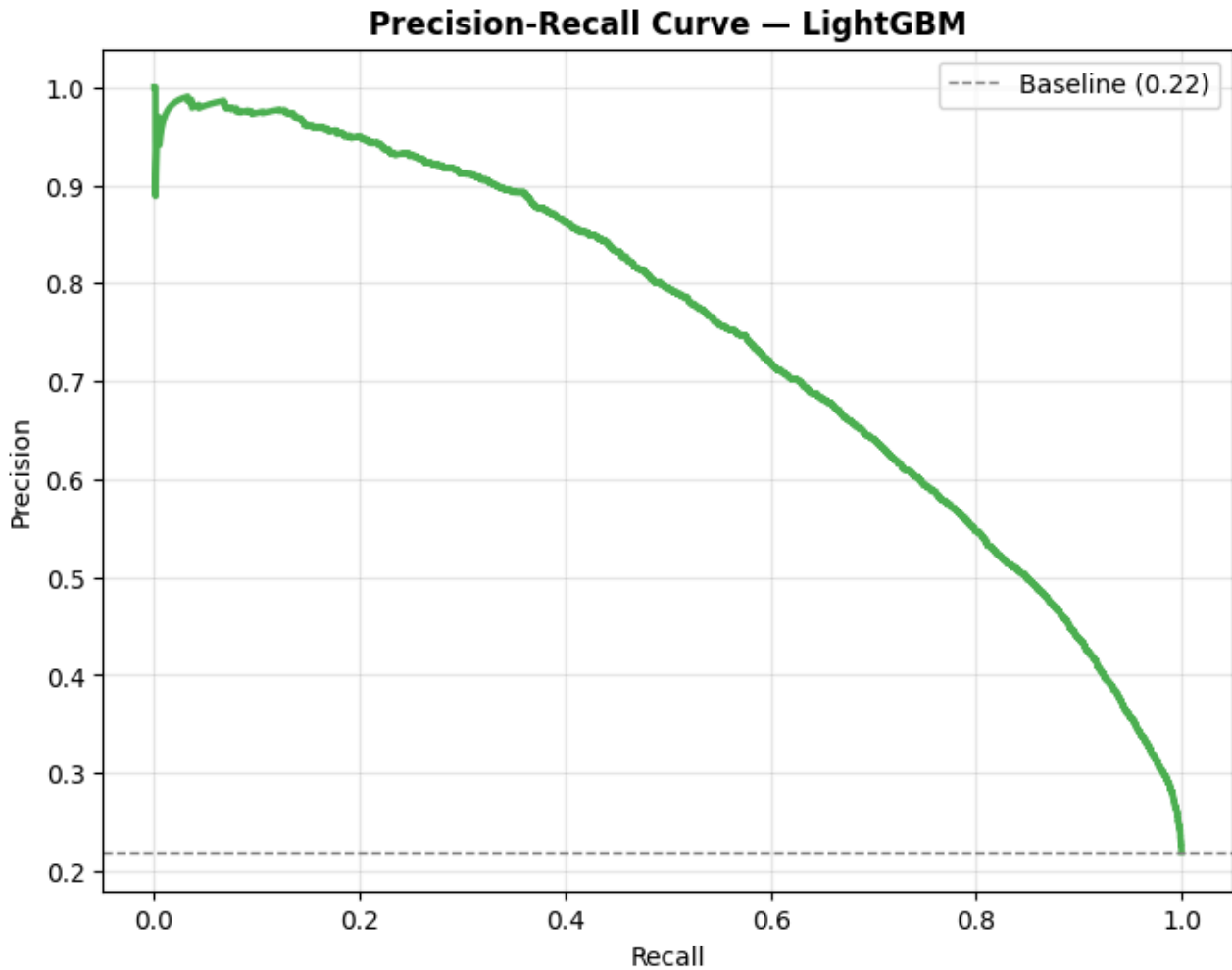
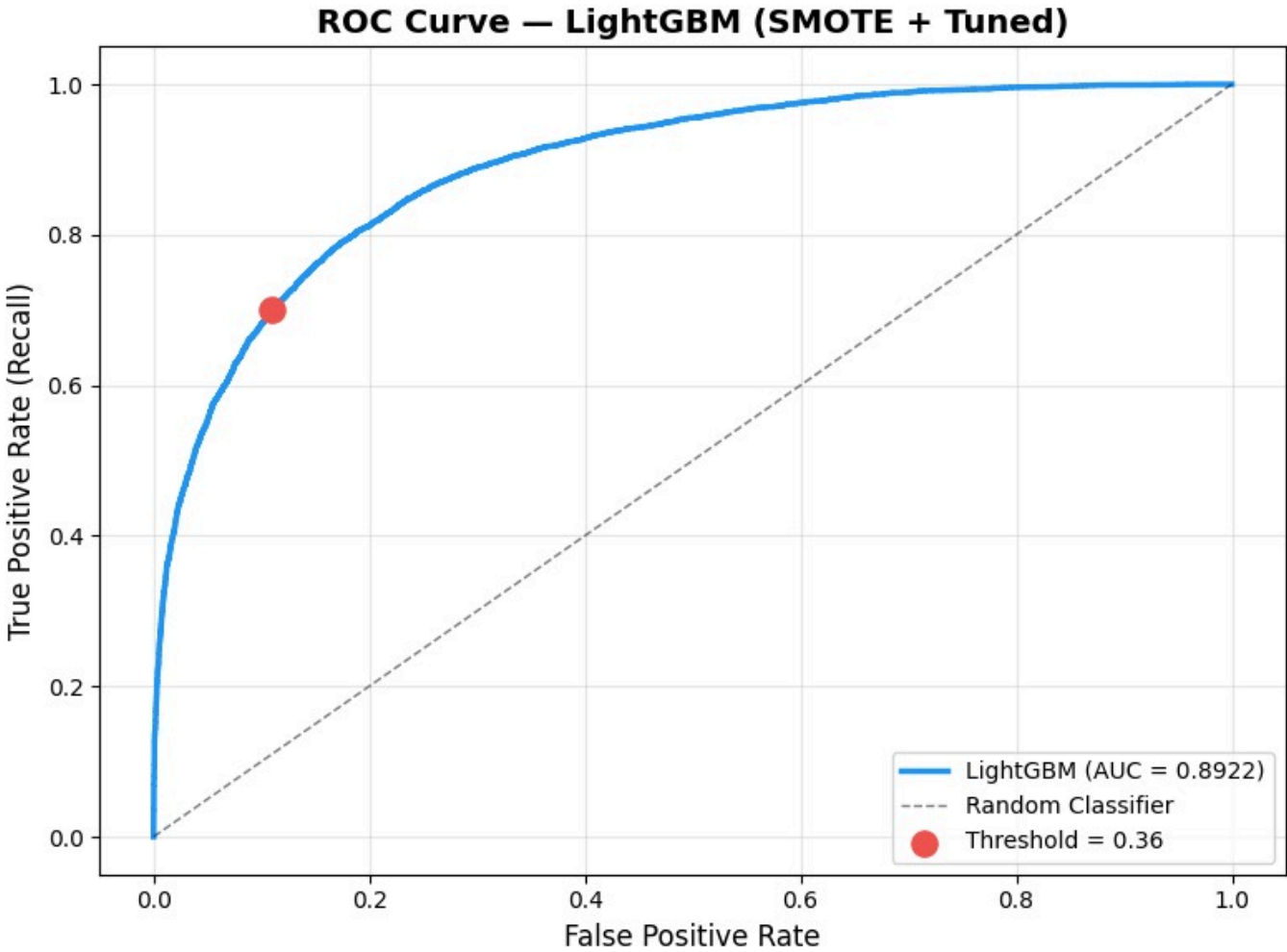


GLOBAL PERFORMANCE

	PRECISION	RECALL	F1-SCORE
0	0.91	0.89	0.90
1	0.64	0.70	0.67

SUPPORT
22717
6375

ACCURACY
85%



SUPPORT VECTOR MACHINE (SVM)

PREDICTIVE ML MODEL



WHAT IS AN SVM?

MAXIMUM MARGIN CLASSIFIER

A supervised learning algorithm that finds the optimal hyperplane separating classes by maximizing the margin between the nearest data points (support vectors).

WHY WE CHOSE IT?

EFFECTIVENESS ON SMALL DATA

Its ability to perform well in high-dimensional feature spaces with relatively small training sets makes it a strong complement to tree-based models.

HOW DOES IT WORK?

KERNEL TRICK

The SVM maps input features into a higher-dimensional space using a kernel function (RBF, polynomial, or linear). It then finds the hyperplane that maximises the margin between classes, ignoring misclassified points through the soft-margin penalty parameter C .

CONNECTION TO OUR DATASET

HANDLES CLASS IMBALANCE

Weather datasets are often imbalanced (rain is less frequent). SVM's margin-based approach and `class_weight` parameter allow it to correctly penalise misclassification of minority rain events.

SUPPORT VECTOR MACHINE (SVM) IMPLEMENTATION



DATA PREPARATION & CLASS BALANCING

Split: 80/20 stratified split with **random_state=42** — preserves rain/no-rain ratio.

SMOTE: Oversampled minority class from 3.56:1 → 1:1 on training set only. Test set untouched.

PCA: Reduced dimensions retaining 95% variance (**n_components=0.95**) to speed up LinearSVC.

MODEL ARCHITECTURE & PROBABILITY CALIBRATION

Core: **LinearSVC** with **C=1.0**, **class_weight='balanced'**, **max_iter=5000**.

Calibration: **CalibratedClassifierCV** wraps LinearSVC to produce probabilities (Platt scaling).

Threshold: Custom decision threshold set at **0.45** to boost recall on rain events.

CROSS-VALIDATION & EVALUATION

CV: 5-fold **StratifiedKFold** on PCA-transformed SMOTE data — accuracy scores averaged.

Metrics: Accuracy · Precision · Recall · F1 + Confusion Matrix on held-out test set.

Saved: **joblib.dump(svm_model, 'svm_model.pkl')** — full pipeline persisted.

SUPPORT VECTOR MACHINE (SVM)

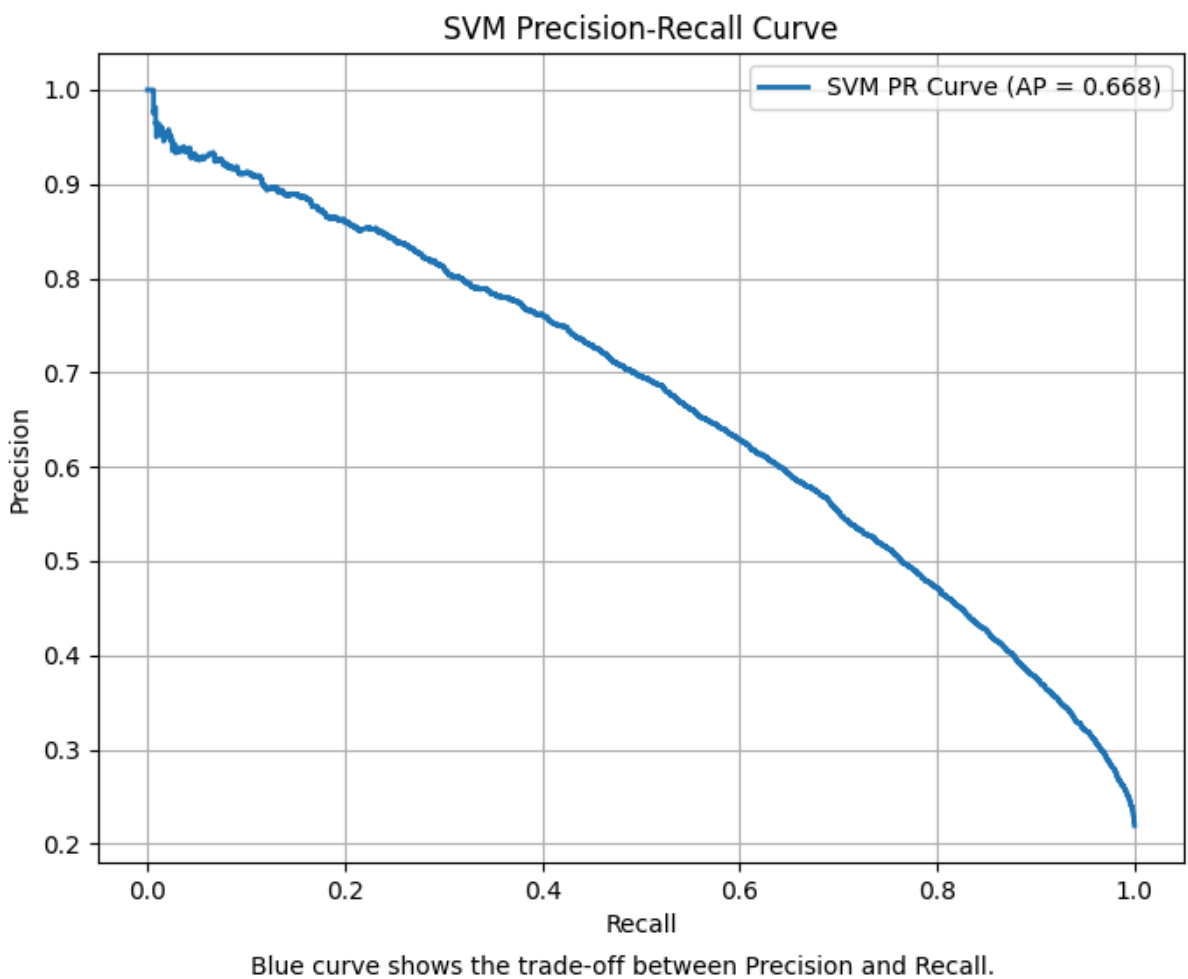
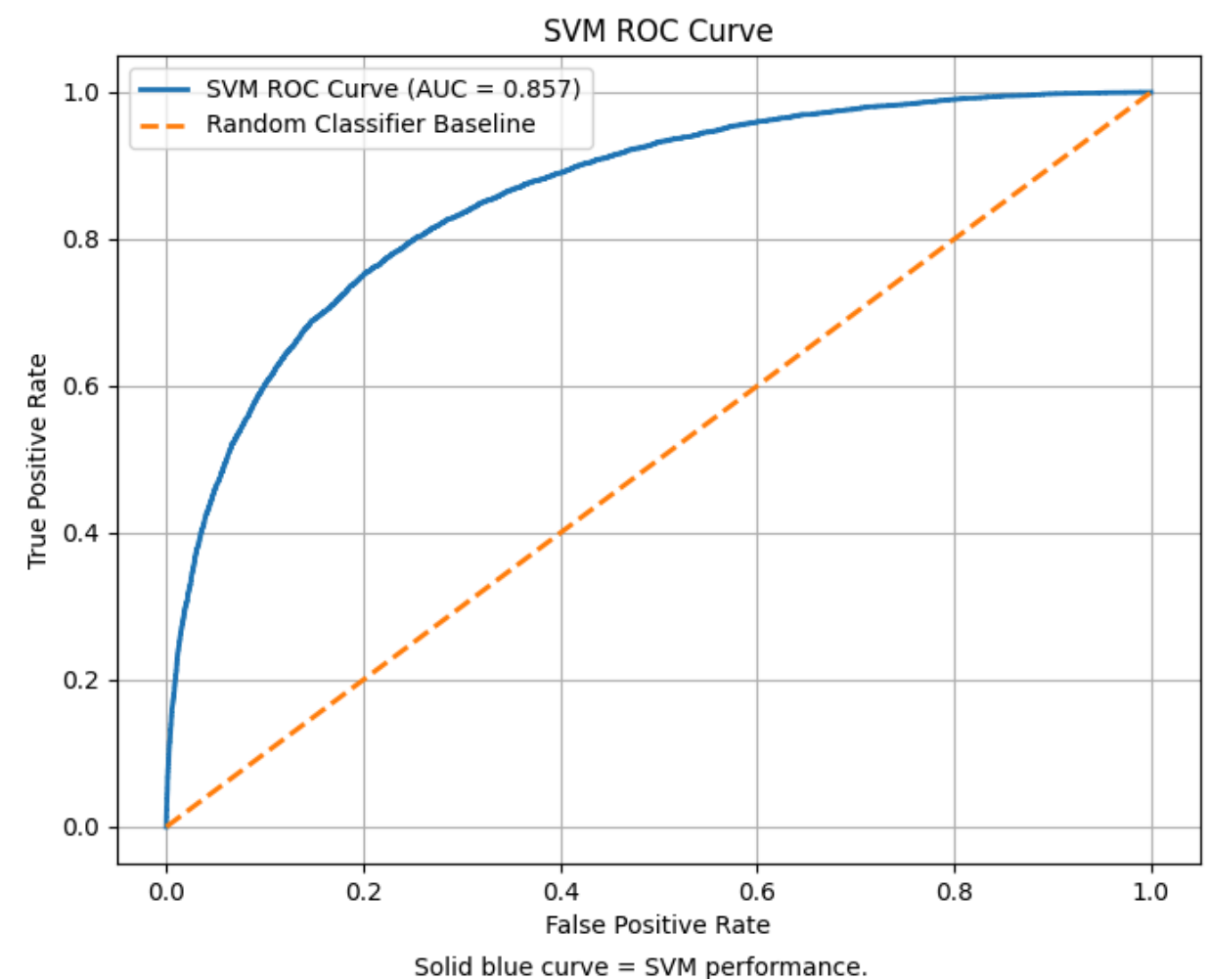
PREDICTIVE ML MODEL



GLOBAL PERFORMANCE

	PRECISION	RECALL	F1-SCORE	SUPPORT
0	0.93	0.74	0.82	22717
1	0.46	0.81	0.59	6375

ACCURACY
75 %



GATED RECURRENT UNIT (GRU)

PREDICTIVE DL MODEL



WHAT IS A GRU?

RECURRENT LEARNING

A type of recurrent neural network (RNN) that uses gating mechanisms to selectively retain or discard information across time steps, enabling sequence modelling.

WHY WE CHOSE IT?

TEMPORAL MEMORY

Weather is inherently sequential. The GRU's ability to learn long-range dependencies in time series data makes it ideal to capture multi-day meteorological patterns preceding rainfall.

HOW DOES IT WORK?

RESET & UPDATE GATES

Two learned gates control the hidden state at each time step: the reset gate decides how much past information to forget, while the update gate controls how much new information flows in balancing memory and adaptability.

CONNECTION TO OUR DATASET

MODELS WEATHER SEQUENCES

By feeding consecutive daily records as input sequences, the GRU learns recurring atmospheric patterns – such as pressure drops followed by humidity spikes – that reliably precede rainfall events.

ARCHITECTURE AT A GLANCE

Input shape	(N, 1, feats)
GRU units	32 or 64
Dropout	0.2 or 0.3
Dense	32→ReLU→Sig
Loss	BinaryXEntropy
Optimizer	Adam
Batch size	32
Early stop	patience=3/5

DATA RESHAPING

Raw tabular data is reshaped to (**samples, 1, features**) — a 3-D tensor required by Keras recurrent layers. Class weights are computed via **compute_class_weight** to handle the rain/no-rain imbalance without SMOTE.

HYPERPARAMETER SEARCH

Grid search across **units** $\in \{32, 64\}$ and **dropout** $\in \{0.2, 0.3\}$. Each config trained for 10 epochs with **EarlyStopping(patience=3)** and 20% validation split. Best config selected by highest val_accuracy.

FINAL TRAINING

Best hyperparams retrained with **EarlyStopping(patience=5)**, **class_weight** applied to penalise missed rain events.

EVALUATION

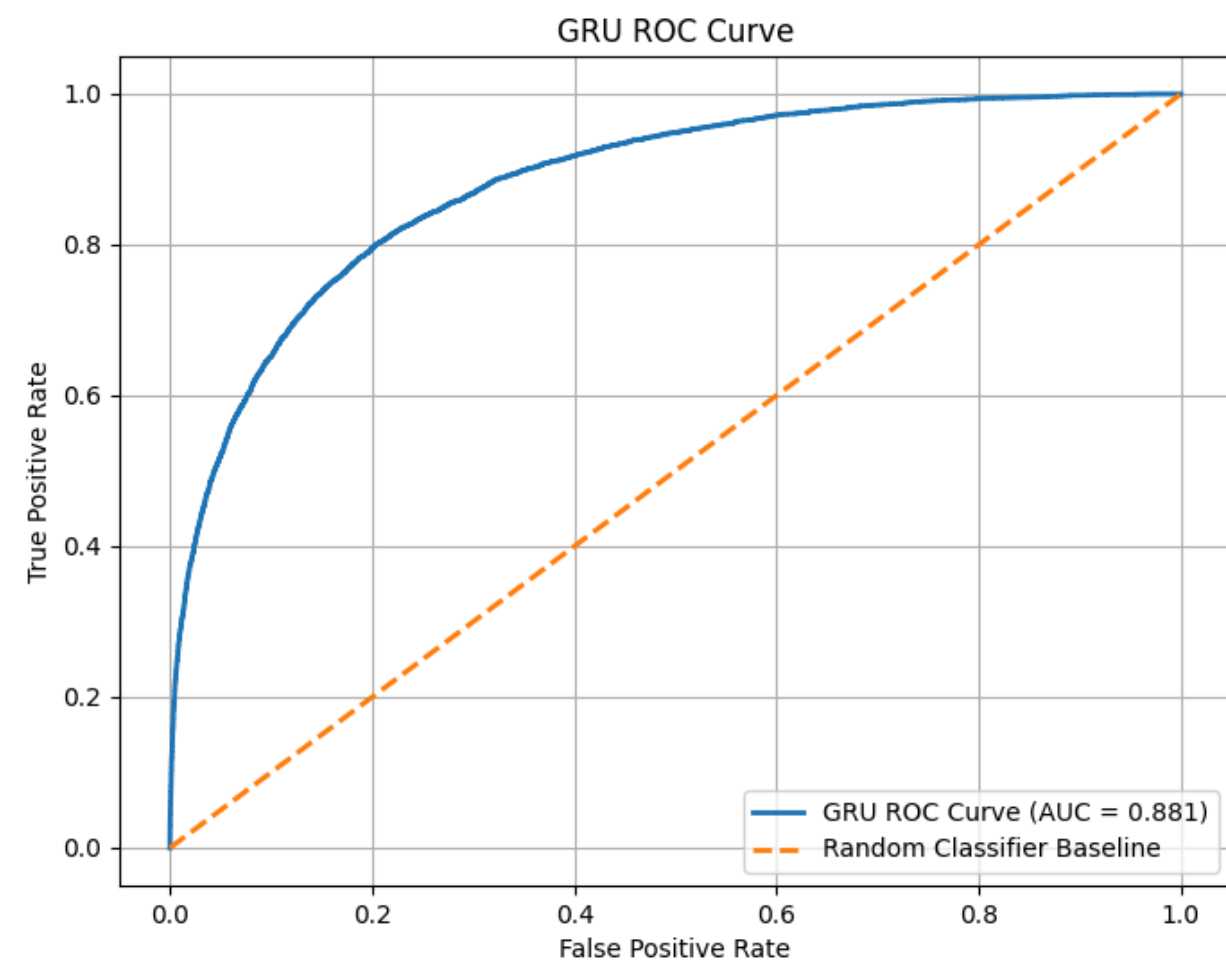
Predictions thresholded at **0.5**. Metrics: Accuracy · Precision · Recall · **F1**. Final model evaluated on the untouched test split alongside SVM for direct comparison.

GLOBAL PERFORMANCE

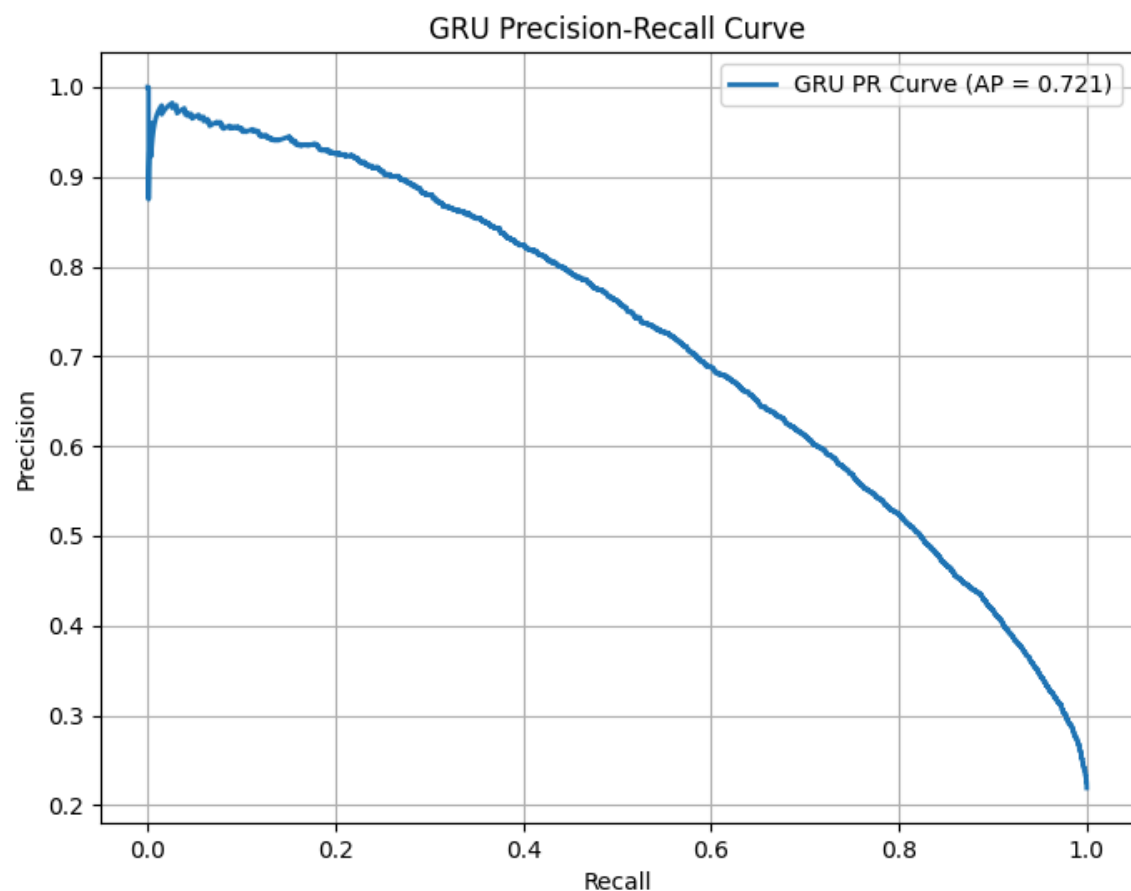
	PRECISION	RECALL	F1-SCORE	SUPPORT
0	0.93	0.81	0.87	22717
1	0.53	0.78	0.64	6375

ACCURACY

80 %



Solid blue curve = GRU model performance.

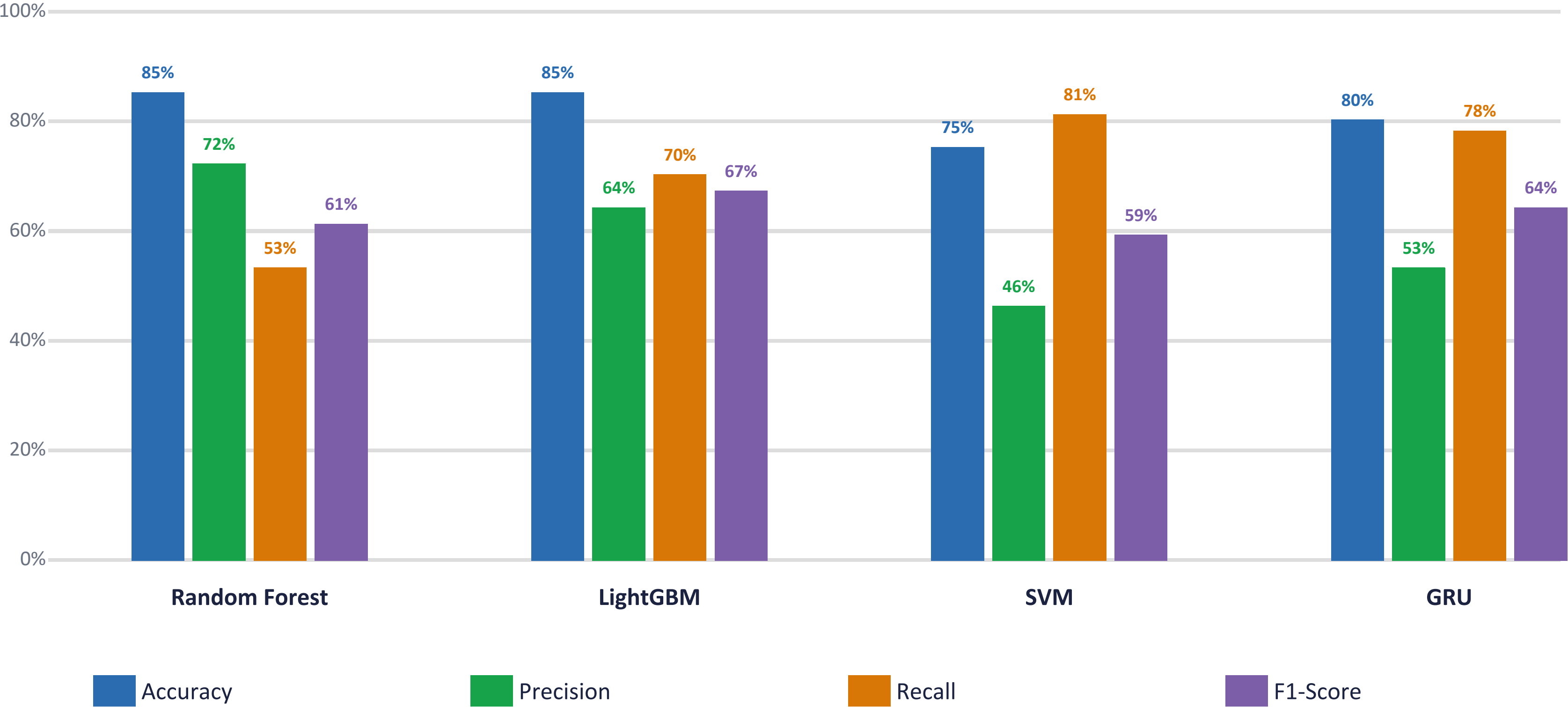


RESULTS INTERPRETATION & ANALYSIS

UNDERSTANDING OUR RESULTS



All metrics below are for the Rain class (class 1) — the minority and most critical class for weather prediction.



RESULTS INTERPRETATION & ANALYSIS

UNDERSTANDING OUR RESULTS



RANDOM FOREST

BASELINE ENSEMBLE

BEST PRECISION

85%

ACC

0.61

F1

0.886

AUC

0.53

RECALL

- ✓ Highest precision (0.72) — fewest false alarms
- ✓ AUC 0.886 — excellent probability ranking
- ✓ Platt Scaling fixes probability distortion

Ideal when false alarms are costly (e.g. aviation, event planning).

LIGHTGBM

GRADIENT BOOSTING

BEST OVERALL

85%

ACC

0.67

F1

0.892

AUC

0.70

RECALL

- ✓ Best F1 (0.67) — optimal precision/recall balance
- ✓ Highest AUC 0.892 — best class separation
- ✓ SMOTE + threshold tuning maximises rain detection

Best for general forecasting where missing rain is expensive.

SVM

LINEAR KERNEL + PCA

BEST RECALL

75%

ACC

0.59

F1

0.857

AUC

0.81

RECALL

- ✓ Highest Rain recall (0.81) — catches 4 in 5 rain events
- ✓ PCA reduces overfitting in high-dimensional space
- ✓ Threshold 0.45 optimised for minority class sensitivity

Best when missing rain is critical (flood warnings, agriculture).

GRU

DEEP LEARNING

BEST DEEP MODEL

80%

ACC

0.64

F1

0.881

AUC

0.78

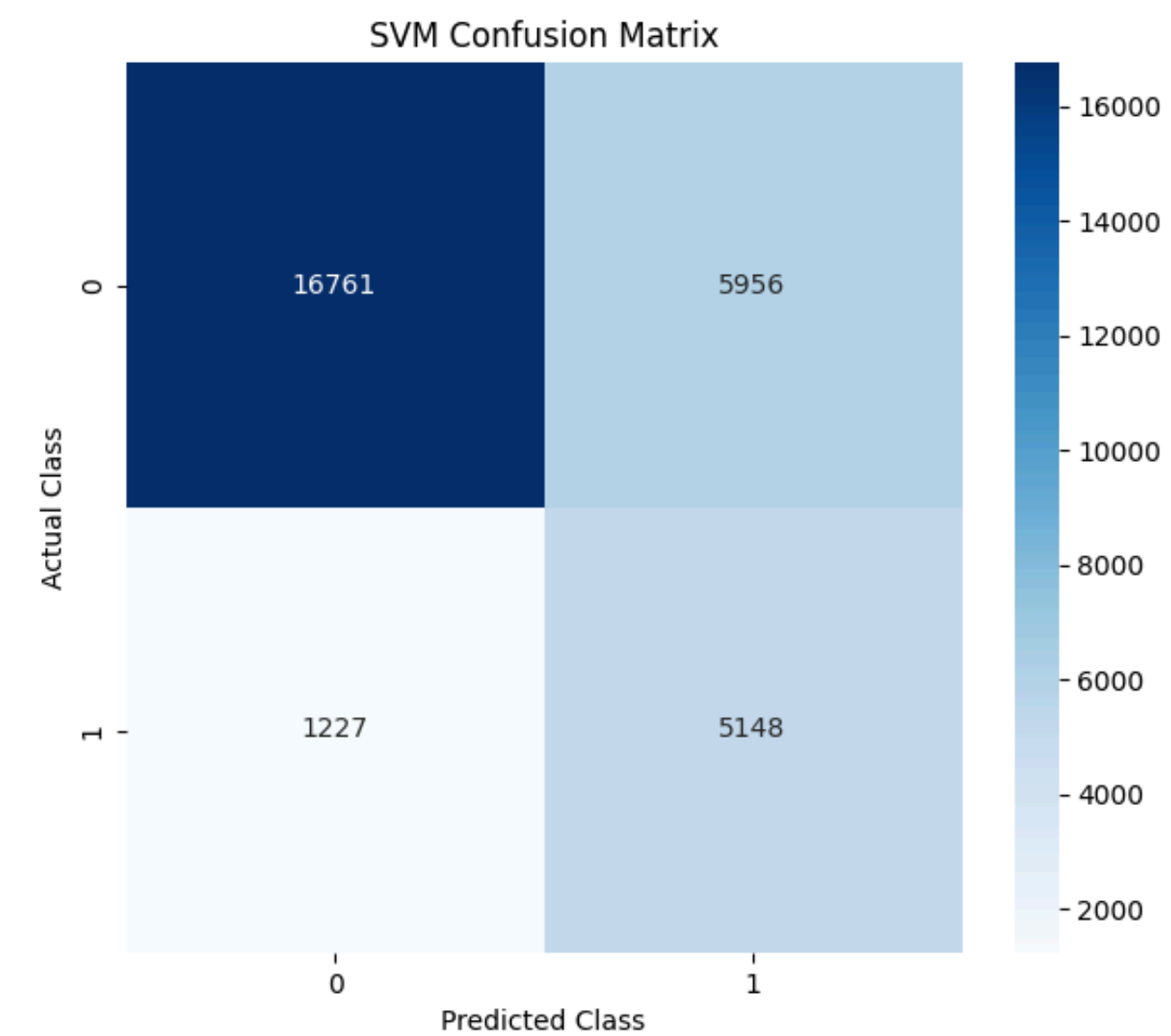
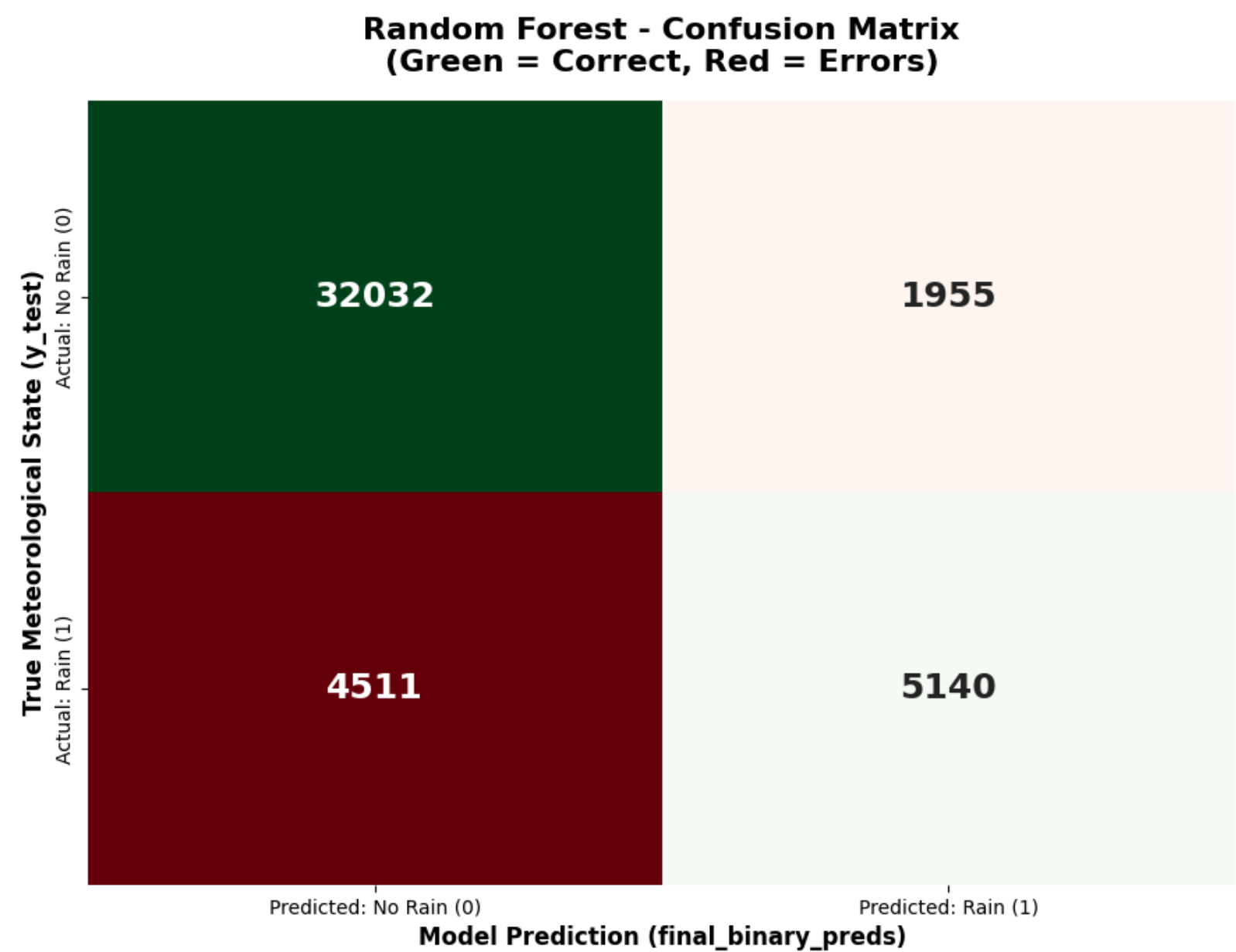
RECALL

- ✓ Strong recall (0.78) with good F1 — balanced deep baseline
- ✓ No SMOTE needed — class weights handle imbalance natively
- ✓ Architecture generalises: can be extended with more time steps

Highest ceiling: with multi-day sequences, expected to surpass RF/LightGBM.

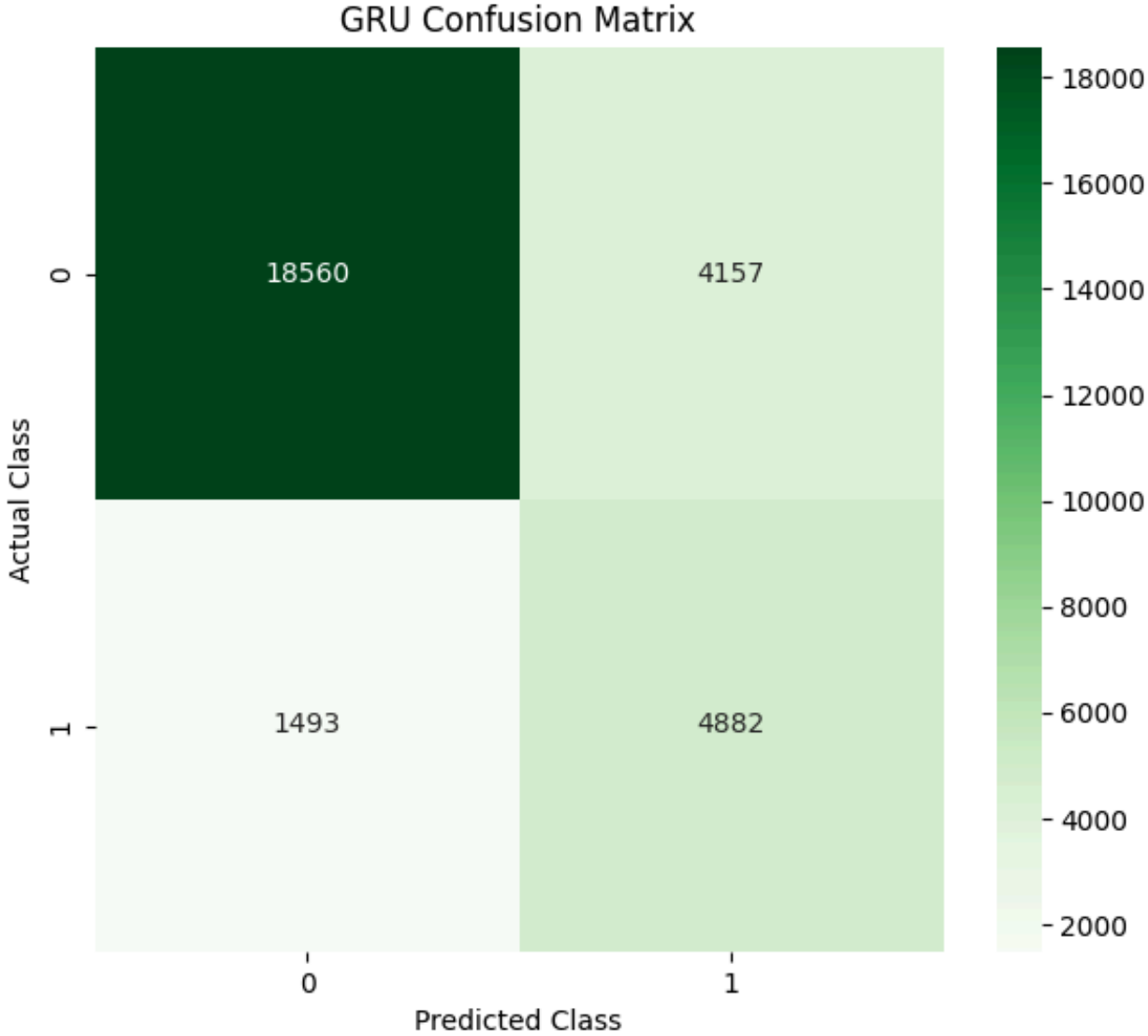
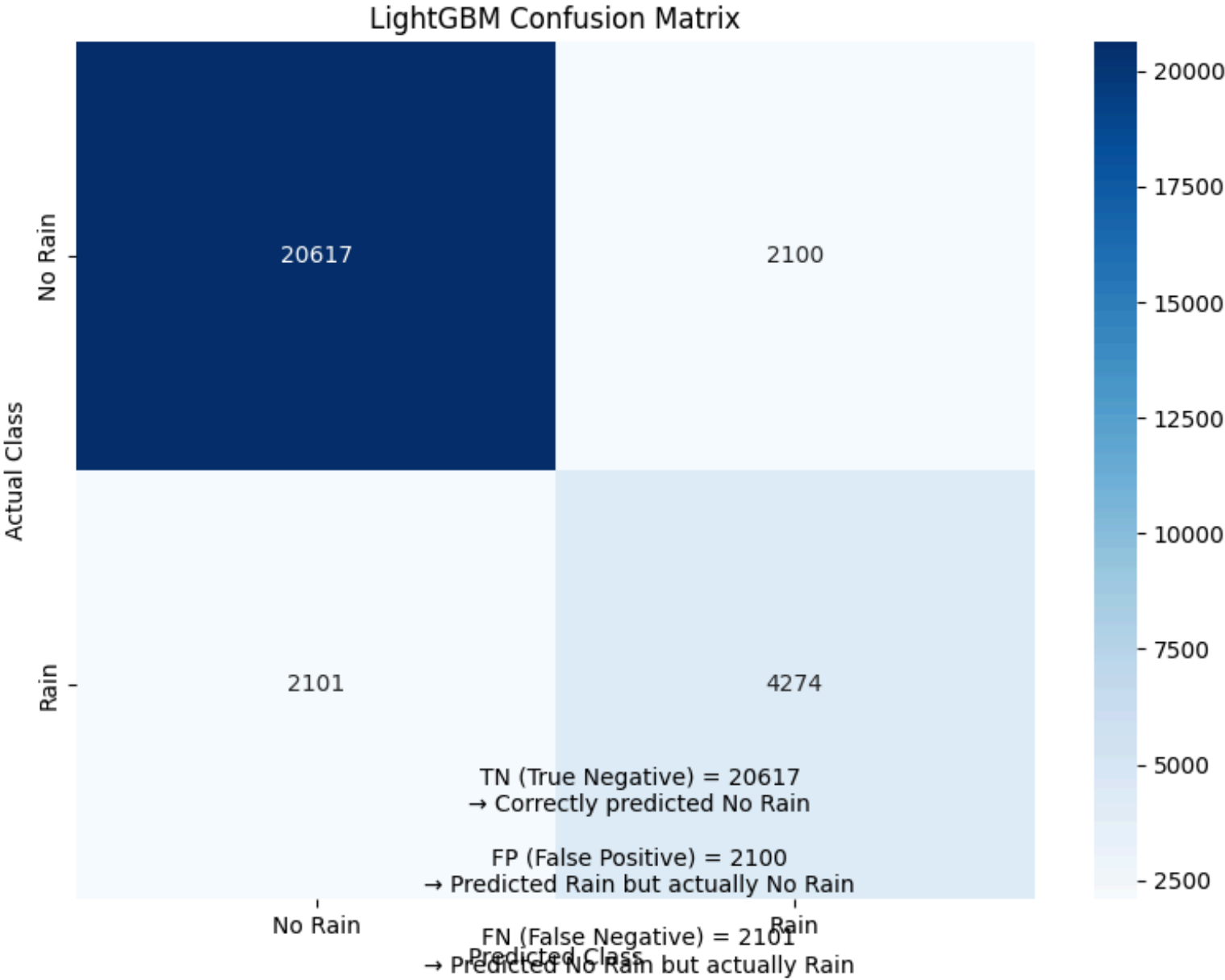
RESULTS INTERPRETATION & ANALYSIS

UNDERSTANDING OUR RESULTS



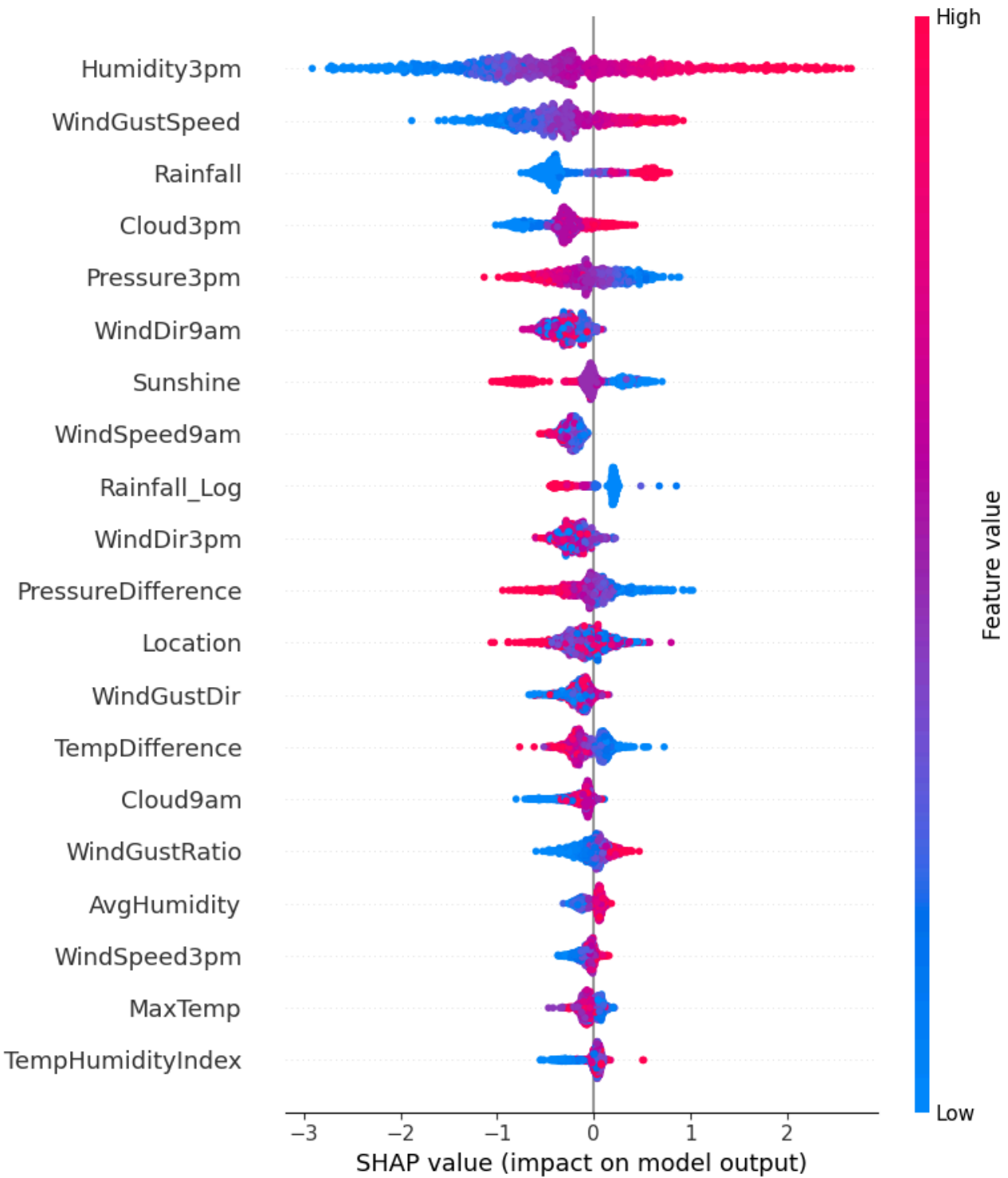
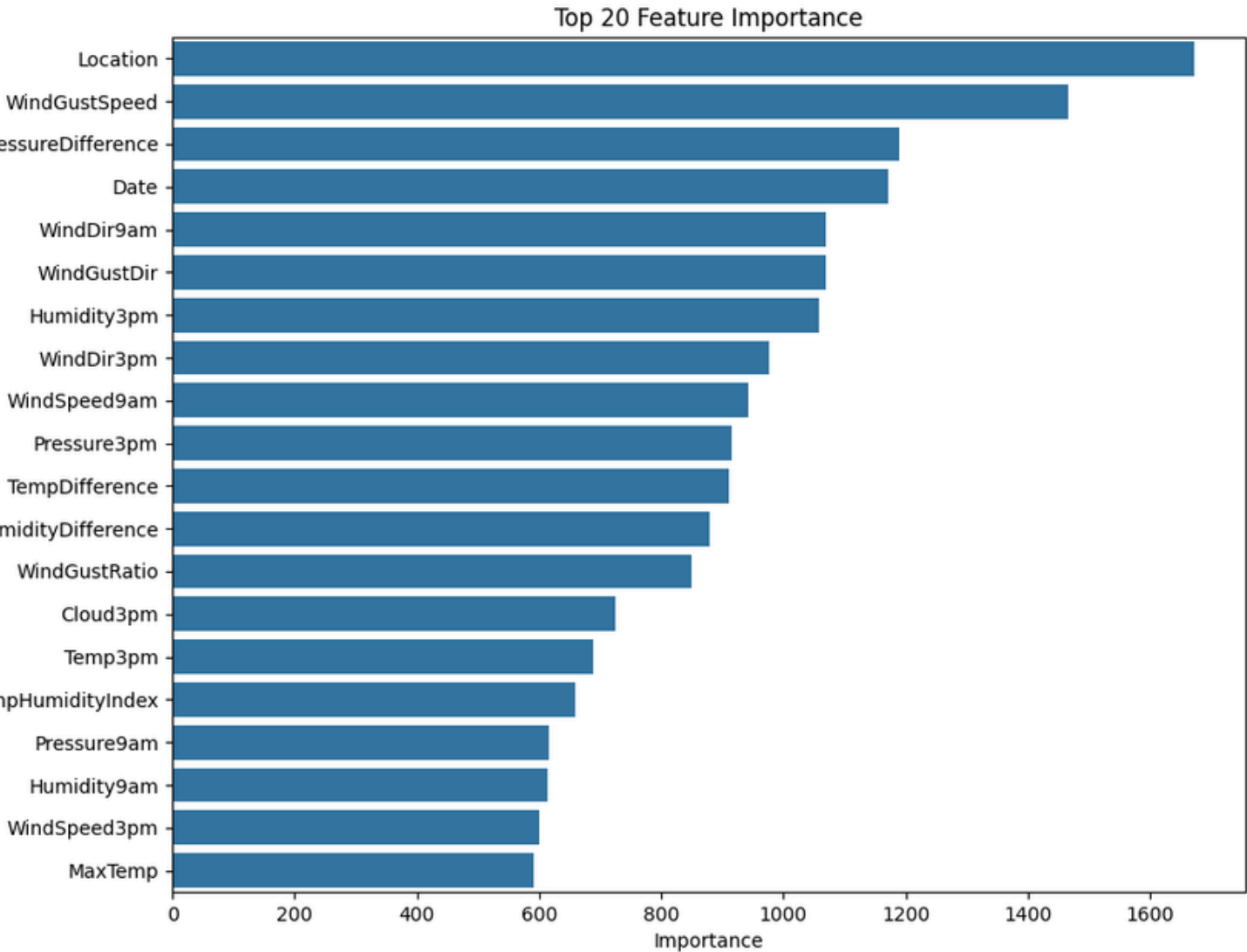
RESULTS INTERPRETATION & ANALYSIS

UNDERSTANDING OUR RESULTS



RESULTS INTERPRETATION & ANALYSIS

FEATURE IMPORTANCE & SHAP LIGHTGBM



ROBUST END-TO-END PIPELINE

Successfully processed **145k+ records**, tackling extreme class imbalance (**SMOTE**) and erratic outliers.

METEOROLOGICAL FEATURE ENGINEERING

Custom physical indicators (e.g., **Pressure-Humidity interactions**) proved crucial in capturing **non-linear** storm triggers.

ALGORITHM EVOLUTION

Evolved from a calibrated Random Forest baseline, moving to SVM and concluding to advanced Gradient Boosting (LightGBM) and Deep Learning (GRU) to maximize **Recall**.

REAL-WORLD APPLICATION

Deployed an interactive dashboard allowing users to input live weather metrics and test "**what-if**" scenarios in real-time.



IPS PROGRAM SPRING 2026

Thank you!

A white feather with a blue shaft and blue dots on its shaft, positioned to the right of the 'Thank you!' text.

YOUSSEF ALY - ISELA ALLHASA - PIETRO VISENTIN